

Multi-omics investigation of spontaneous T2DM macaque reveals gut microbiota promote T2DM by up-regulating the absorption of excess palmitic acid

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Xu Liu, Shengzhi Yang, Yuchen Xie, Cong Jiang, Ke Shang, Jinxia Luo, Lin Zhang, Gang Hu, Qinghua Liu, Bisong Yue, Zhenxin Fan, Zhanlong He, Jing Li 

Key Laboratory of Bio-resources and Eco-environment (Ministry of Education), College of Life Sciences, Sichuan University, Chengdu, China • Sichuan Key Laboratory of Development and Application of Monkey Models for Human Major Disease, Sichuan, China • SCU-SGHB Joint Laboratory on Non-human Primates Research, Meishan, Sichuan, China • Institute of Medical Biology, Chinese Academy of Medical Sciences and Peking Union Medical College, Yunnan, China

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eLife Assessment

This **important** work substantially advances our understanding of the interaction among gut microbiota, lipid metabolism, and the host in type 2 diabetes. The evidence supporting the claims of the authors is **solid**, although additional experiments for the control FMT are not yet satisfactory. The work will be of interest to medical biologists working on microbiota and diabetes.

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Abstract

Although gut microbiota and lipid metabolites have been suggested to be closely associated with type 2 diabetes mellitus (T2DM), the interactions between gut microbiota, lipid metabolites and the host in T2DM development remains unclear. Rhesus macaques may be the best animal model to investigate these relationships given their spontaneous development of T2DM. We identified eight spontaneous T2DM macaques and conducted a comprehensive study investigating the relationships using multi-omics sequencing technology. Our results from 16S rRNA, metagenome, metabolome and transcriptome analyses identified that gut microbiota imbalance, tryptophan metabolism and fatty acid β oxidation disorders, long-chain fatty acid (LCFA) accumulation, and inflammation occurred in T2DM macaques. We verified the accumulation of palmitic acid (PA) and activation of inflammation in T2DM macaques. Importantly, mice transplanted with spontaneous T2DM macaque fecal microbiota and fed a high PA diet developed prediabetes within 120 days. We determined that gut microbiota mediated the absorption of excess PA in the ileum, resulting in the accumulation of PA in the serum consequently leading to T2DM in mice. In particular,

we demonstrated that the specific microbiota composition was probably involved in the process. This study provides new insight into interactions between microbiota and metabolites and confirms causative effect of gut microbiota on T2DM development.

Introduction

Diabetes mellitus is considered to be a refractory disease causing a significant socio-economic burden. Type 2 diabetes mellitus (T2DM) is the dominant type of diabetes mellitus characterized by metabolic disorders, insulin resistance, and deficiency of insulin secretion (1, 2). The pathogenesis of this chronic disease is complex and genetic and environmental factors, such as sugar and lipid intake, gut microbiota, many metabolites, and even air pollutants, contribute to its increase in prevalence (3, 4). Accumulating evidence has linked gut microbiota with T2DM development by variety ways. The gut microbiota can impact the integrity of the intestinal epithelial barrier, mediate insulin resistance, as well as regulate the function of mitochondria (5, 6, 7). They can regulate local or systemic immunity and inflammation, which also contributes to the development of T2DM (8). Moreover, various gut microbial metabolites, such as short-chain fatty acids, bile acid, and tryptophan-derived metabolites, have been reported to be closely related to the pathogenesis of T2DM (5, 6, 9, 10). However, the interactions between gut microbiota and its host with T2DM have not yet been fully characterized. In pathological conditions, the dysregulation of the host can lead to changes in gut microbiota composition. In turn, the microbiota plays a regulatory role to participate in the development of T2DM. Despite the complicate interactions between host and microbiota in the context of T2DM, some studies suggest antidiabetic interventions targeting the gut microbiota such as fecal microbiota transplanting (FMT) can be applied as a clinical treatment of T2DM (11, 12, 13).

Meanwhile, dysfunction of lipid metabolism contributes to T2DM development by inducing lipotoxicity in humans and animal models (14, 15). The long-chain fatty acids (LCFAs) are the principal lipid components naturally occurring in animal fats and vegetable oil, as well as the main metabolites of fat. LCFAs such as palmitic acid (PA, C16:0), palmitoleic acid (C16:1N7), and oleic acid (C18:1N9), are reported to have strong association with T2DM. In the last few decades, there is increasing evidence that frequent consumption of LCFAs contributes to metabolic diseases such as obesity and T2DM due to the high PA content (16, 17). PA is a saturated fatty acid and its increase in serum is a significant contributing factor in T2DM development (18, 19, 20). The suggested mechanisms by which PA mediates T2DM include increasing diacylglycerol and ceramide synthesis (21), mitochondrial and endoplasmic reticulum stress (22, 23), and activation of pro-inflammatory pathways (24). Nevertheless, whether gut microbiota involving in PA mediated T2DM and the interactions between gut microbiota and LCFA metabolites in T2DM development are still unclear.

Spontaneous development of T2DM in non-human primates (e.g. macaques) is highly similar to human T2DM, such as insulin resistance in early stages and later abnormal glucose tolerance and T2DM development follows the same pathological changes of pancreatic islets and complications. In fact, T2DM macaques avoid medication interference and environmental heterogeneity under controlled experimental conditions, and share key pathological features with humans, such as amyloidosis of pancreatic islets, which is absent in mouse models (25, 26), suggesting that T2DM macaques are the optimal animal model for simulating human T2DM and its complications (27). However, previous studies indicated that naturally occurring spontaneous T2DM macaques in captive populations were rare even if individuals were given a high lipid and high sugar diet (28, 29). After nasally fed cynomolgus macaques with a high-fat dietary emulsion for 12 months, the macaques did not experience significant increases in fasting blood glucose and glycosylated hemoglobin (29). The controversial effects of high fat on the T2DM development are worthy of further investigations to better understand this complex disease.

Here, this study identified spontaneous development of T2DM in individuals (hereafter spontaneous T2DM macaques) from a large group of rhesus macaques. These spontaneous T2DM macaques have never been treated with anti-diabetic drugs and therefore provide valuable models for pathogenesis investigation of T2DM. Based on the macaque model, we used multi-omics techniques to address the interactions between gut microbiota, host gene expression, and fecal metabolites and the development of T2DM. Our results demonstrated that gut microbiota and LCFA metabolites played important roles in the pathogenesis of T2DM. We validated the increased content of plasma PA and activation of inflammation in the T2DM macaques. In addition, we successfully induced prediabetes in mice by transplanting fecal microbiota from T2DM macaques into mice in conjunction with a diet high in PA. We also revealed the specific structure of gut microbiota that promoted T2DM development by regulating the absorption of excess PA, which provides key evidence of the causative effect of gut microbiota on the pathogenesis of T2DM.

Results

Metagenome and 16S sequencing demonstrate alterations on gut microbiota in spontaneous T2DM macaques

We identified eight spontaneous T2DM macaques out of 1698 individuals from long-term glucose monitoring in a captive population (Table S1). The fasting plasma glucose (FPG), fasting plasma insulin (FPI), and HOMA-IR levels in the T2DM macaques were significantly higher than in the control group ($p < 0.01$), suggesting insulin resistance in T2DM macaques. However, glycosylated hemoglobin A1c (HbA1c), triglycerides (TG), total cholesterol (TC), high-density lipoprotein cholesterol (HDL), low-density lipoprotein cholesterol (LDL), and BMI did not significantly differ from the controls ($p > 0.05$, **Table 1**). Comparison of gut microbiota between the two groups based on 16S rRNA amplicon found that both Shannon and Simpson index decreased in T2DM macaques but it was not significantly different from controls ($p > 0.05$; **Figures 1A** and **B**). PCoA indicated that the microbiota composition of spontaneous T2DM macaques was different from the control group (**Figure 1C**). As one of the most dominant families in both groups, Lachnospiraceae showed significantly higher abundance in the T2DM group (14.48%) than in the control group (6.66%). While the abundance of the Lactobacillaceae family was significantly greater in the control group (25.95%) compared to spontaneous T2DM macaques (20.88%) (**Figure 1D**). A total of 21 microbes were identified as differential microbes between the T2DM group and control group, where ten microbes including Erysipelotrichaceae and five members in Lachnospiraceae family (*Ruminococcus gnavus* (current name: *Mediterraneibacter gnavus*), *Lachnospira*, *Coprococcus* sp. *Dorea longicatena*, and *Roseburia*) were significantly greater in the T2DM group (**Figure 1E**).

The metagenome results showed that the T2DM group had a higher abundance of Erysipelotrichaceae, *Eubacterium rectale*, Lachnospiraceae, Negativicutes, *Blautia*, and Coriobacteriia than the control group (Figure S1A). Functional enrichment demonstrated a total of 74 KEGG pathways with significant differences between the two groups. These pathways were mainly associated with T2DM and inflammation, including type II diabetes mellitus, glucagon signaling pathway, starch and sucrose metabolism, lipid metabolism, and toll-like receptor signaling pathways (**Figure 1F**). Among the six CAZy enzyme families, the Carbohydrate-binding module (CBMs), Glycosyl Transferases (GTs), Carbohydrate esterases (CEs), and Polysaccharide Lyases (PLs) families were significantly upregulated in the T2DM group, indicating significant changes in carbohydrate metabolism (**Figure 1G**). The results of metagenome and 16S sequencing demonstrated significant alterations in the composition and function of gut microbiota in spontaneous T2DM macaques, with a greater abundance of microbes associated with T2DM and fewer beneficial microbes.

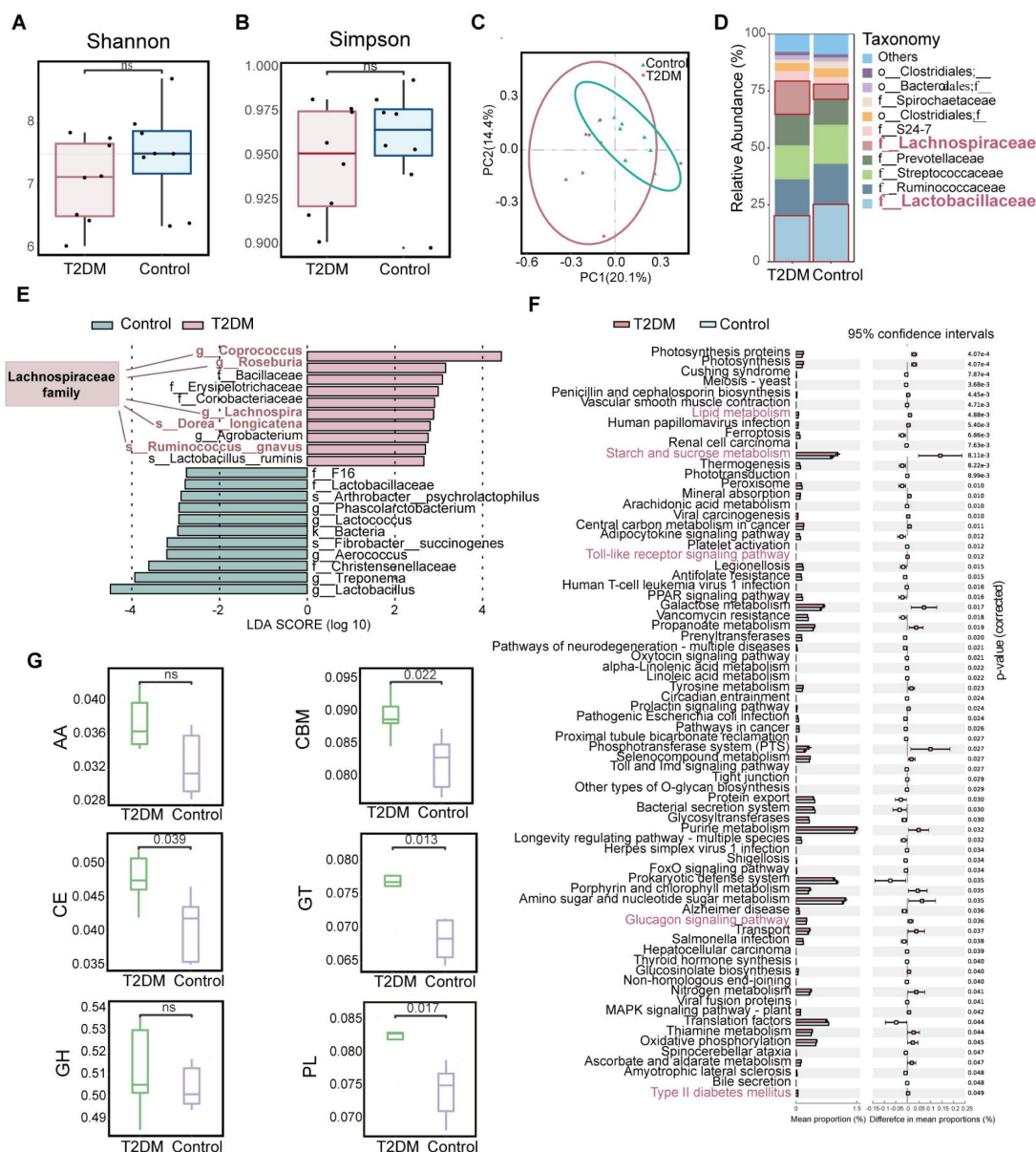


Figure 1.

The changes in gut microbiota in spontaneous T2DM macaques

(A) Alpha diversity estimates (Shannon index) between T2DM and control groups (ns, not significant, two-tailed t-test, $n=8$).

(B) Alpha diversity estimates (Simpson index) between T2DM and control groups (ns, not significant, two-tailed t-test, $n=8$).

(C) Principal Coordinate Analysis (PCoA) ($n=8$).

(D) Differential analysis of gut microbial composition in T2DM and control groups ($n=8$).

(E) LEfSe analysis between T2DM and control groups ($n=8$).

(F) Differential analysis of gut microbial function in T2DM and control groups ($n=5$). The pathways with red color were associated with T2DM and inflammation. Error bar is mean with $\pm$ standard deviation (s.d.).

(G) Differential analysis of gut microbial CAZy enzyme in T2DM and control groups ($n=5$). CBMs: carbohydrate-binding module ($p=0.022$, two-tailed t-test); GTs: Glycosyl Transferases ($p=0.013$, two-tailed t-test); PLs: Polysaccharide Lyases ($p=0.017$, two-tailed t-test); AA: Auxiliary activity enzymes (ns, not significant, two-tailed t-test); GH: Glycoside hydrolases (ns, not significant, two-tailed t-test); CE: Carbohydrate esterases ($p=0.039$, two-tailed t-test). For all boxplots: centre lines, upper and lower bounds show median values, 25th and 75th quantiles; upper and lower whiskers show the largest and smallest non-outlier values. In c, ellipses represent the 95% confidence intervals.

Index	T2DM (n=8)	Control (n=8)
FPG (mmol/L)	7.75±1	4.05±1.02 ^{**}
FPI (μU/mL)	17.97±8.50	6.45±2.56 ^{**}
HOMA-IR	6.24±3.06	1.20±0.65 ^{**}
BMI	14.03±7.71	14.30±1.42
HbA1c (%)	3.78±0.70	3.26±0.62
TG (mmol/L)	0.97±0.53	0.68±0.42
TC (mmol/L)	3.30±0.98	3.52±0.86
HDL (mmol/L)	1.23±0.40	1.42±0.38
LDL (mmol/L)	1.36±0.44	1.47±0.57

* $p < 0.05$, ** $p < 0.01$

Table 1

Physiological and biochemical parameters of Control and T2DM group

Fecal metabolome and blood transcriptome reveals dysfunction of fatty acid β oxidation and tryptophan metabolism in T2DM macaques

The UHPLC-MS based metabolome analysis on fecal samples of T2DM and control macaques identified 1564 metabolites belonging to various types of secondary metabolites, with lipids and lipid-like molecules being most abundant (31.1%) (Figure S1B). We found 64 significantly differential metabolites between the two groups using a combined multidimensional statistical analysis (OPLS-DA) and univariate statistical analysis (T-test) ($VIP > 1$, $p < 0.05$) (Figures 2A and B; Table S2). Among them, muscon, indole-3-acetaldehyde, and serotonin were significantly lower in the T2DM group (Figure 2C) and are associated with anti-inflammatory activity (34, 35, 36). Notably, 64 differential metabolites were significantly enriched in one pathway: tryptophan metabolism (Figure 2D), and we identified two significantly different metabolites, indole-3-acetaldehyde and serotonin in this pathway. Both metabolites are microbiota-derived tryptophan metabolites and the ligands for AhR (35, 36). The lower concentration of AhR ligands may lead to the development of inflammation and metabolic syndromes in humans (37). In addition, we found the contents of many acylcarnitine metabolites were significantly higher in the T2DM group ($VIP > 1$, $p < 0.1$), including l-propionylcarnitine, hexanoyl-l-carnitine, (r)-butyrylcarnitine, and isovaleryl-l-carnitine (Table S2). Among them, l-propionylcarnitine, a kind of acylcarnitines, was the most upregulated metabolite in the T2DM macaques (FC: 16.19) (Figure 2C). Acylcarnitines are products of incomplete oxidation of LCFAs, which can activate pro-inflammatory signaling pathways and ultimately inhibit insulin activity (38). Our results indicated incomplete LCFAs β oxidation in spontaneous T2DM macaques, while similar characteristics were also observed in insulin resistant and T2DM humans (39, 40).

Blood transcriptome analysis was consistent with metabolome results, indicating dysfunction of fatty acid β oxidation and inflammation in T2DM macaques. Gene expression in T2DM macaques exhibited significant differences from the controls (Figure 2E), and a total of 161 differentially expressed genes (DEGs) (26 upregulated and 135 downregulated) were identified in T2DM macaques at a FDR level of 0.05 (Figure 2F). Enrichment analysis of the DEGs was linked to diabetes, fatty acid metabolism, and inflammation, such as diabetic cardiomyopathy, glucose homeostasis, fatty acid metabolic process, and chemokine production (Figure 2G). We also identified 26 differential enrichment pathways between the two groups by aggregate fold change (AFC), and most were associated with insulin resistance and inflammation, including insulin resistance, PI3K-Akt signaling pathway, bacterial invasion of epithelial cells, and NOD-like receptor signaling pathway (Figure S2A). As shown in the insulin resistance pathway, expression of *IL6* and *IRS1* were upregulated, while *INSR* was downregulated in T2DM macaques (Figure S2B). The WGCNA analysis identified four modules that were related to T2DM. Darkgreen module and brown module were significantly positively correlated with T2DM, while green module and darkred module were negatively correlated with T2DM (Figure 2H). Genes in the darkgreen module and brown module related to lipolysis and inflammation were significantly upregulated in T2DM macaques, and genes in the green module and darkred module related to fatty acid metabolism and insulin secretion were significantly downregulated (Figure S2C). With the cut-off ($|kME| > 0.8$), a total of 546 genes in the four modules were identified as hub genes (Table S3). Of these 546, 102 hub genes were also DEGs (Figure 2I). Several of these genes have been reported as correlated with T2DM in humans, including *IGF2BP2*, *LEPR*, *RAP1A*, *SESTRIN3*, and *ITLN1* (41, 42, 43, 44, 45). In addition, *ACSM3*, *HADHB*, and *EFHB* are involved in fatty acid β oxidation and inflammation (46, 47, 48).

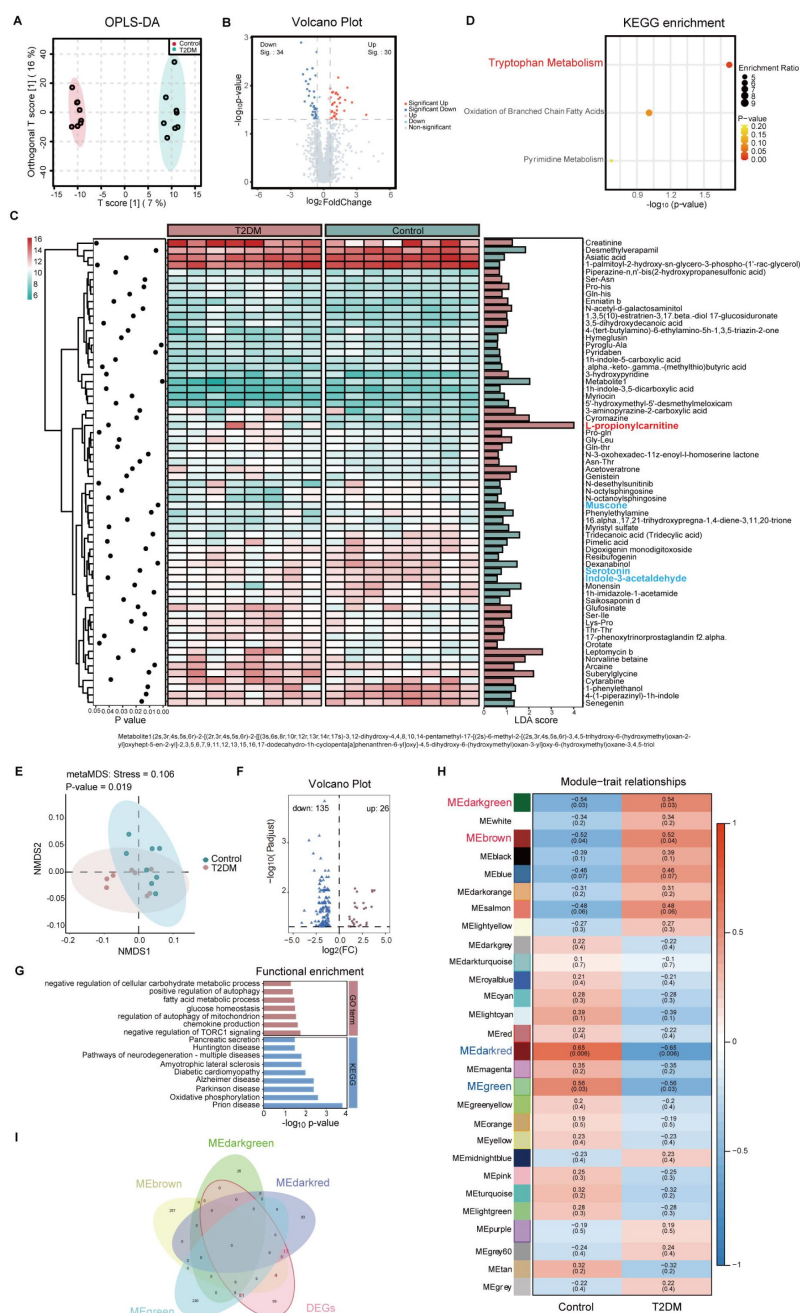


Figure 2.

The alterations of fecal metabolites and gene expression in spontaneous T2DM macaques

- (A) Orthogonal partial least squares discriminant analysis (OPLS-DA) score plots based on the metabolic profiles.
- (B) Volcano plots of metabolomics ($p < 0.05$, two-tailed t-test).
- (C) Fecal metabolites with significant differences between T2DM and control groups ($VIP > 1$, $p < 0.05$, two-tailed t-test).
- (D) Enrichment analysis of the differentially abundant pathways between T2DM and control groups ($p < 0.05$, two-tailed t-test).
- (E) Non-metric multidimensional scaling (NMDS) analysis between T2DM and control groups ($p = 0.019$, two-tailed t-test).
- (F) Volcano plots of DEGs ($\log_2(\text{fold change}) \geq 1$, $p < 0.05$, two-tailed t-test).
- (G) The GO and KEGG pathway enrichment analyses ($p < 0.05$, two-tailed t-test).
- (H) Weighted Gene Co-Expression Network Analysis (WGCNA).
- (I) Venn analysis between hub genes and DEGs. In A and E, ellipses represent the 95% confidence intervals. Data shown are from 8 individuals per group.

Validation of LCFAs accumulation and inflammation in T2DM macaques

Collectively, metabolome and transcriptome results indicated dysfunction in fatty acid β oxidation and tryptophan metabolism in T2DM macaques, which may lead to LCFA accumulation and inflammation. To support this conclusion, we performed targeted medium- and long-chain fatty acid mass spectrometry of plasma and examined serum inflammatory cytokines in the macaques. A total of 34 fatty acids were detected, and among the five types of fatty acids, the concentration of saturated fatty acid (SFA) was significantly greater in the T2DM group ($p < 0.05$, **Figure 3A** [↗](#)), while other types of fatty acids were not significantly different between the two groups ($p > 0.05$, **Figures 3B-E** [↗](#)). In particular, concentrations of PA, palmitoleic acid, and oleic acid were significantly higher in the T2DM group than control group ($p < 0.05$ and $VIP > 1$, **Figures 3F** [↗](#) and **G** [↗](#)). Increased content of PA, palmitoleic acid, and oleic acid was also found in human T2DM ([49](#) [↗](#), [50](#) [↗](#)). PA was an important metabolite mediating insulin resistance through three main mechanisms, being increased diacylglycerol and ceramide synthesis, mitochondrial and endoplasmic reticulum stress, and activation of pro-inflammatory pathways through membrane receptors ([21](#) [↗](#), [22](#) [↗](#), [23](#) [↗](#), [24](#) [↗](#)). Analysis on the serum inflammatory cytokines found that IL-1 β was significantly higher in the T2DM group ($p < 0.05$, **Figure 3H** [↗](#)), but TNF- α and IL-6 levels showed no significant difference between the two groups ($p > 0.05$, **Figures 3I** [↗](#) and **J** [↗](#)). IL-1 β is a major player in a variety of autoinflammatory diseases and a key promoter of T2DM systemic and tissue inflammation ([51](#) [↗](#)). Moreover, blood routine examination showed an increase of white blood cell (WBC) number, neutrophil (NEU) percentage and NEU number and a decrease of lymphocyte (LYM) percentage and LYM number, also indicating the inflammation in the T2DM macaques ($p < 0.05$, **Table 2** [↗](#)).

To investigate effect of gut microbiota on the LCFAs accumulation and inflammation in T2DM macaques, we performed a correlation analysis among the DEGs, differential metabolites, and differential microbes using spearman rank correlation. Four differential microbes in Lachnospiraceae family (*Coprococcus*, *Lachnospira*, *Roseburia* and *Dorea longicatena*) were significantly associated with three differential metabolites of PA, palmitoleic acid and oleic acid ($|r| > 0.5$, adj $p < 0.05$), suggesting the participation of Lachnospiraceae microbes in LCFAs accumulation in T2DM macaques (**Figure 3K** [↗](#)). In addition, the bacteria in class Coriobacteriia was also associated with the three LCFAs ($|r| > 0.5$, adj $p < 0.05$, **Figures S1C** and **D**).

Fecal microbiota transplantation (FMT) with high content PA food induce prediabetes in mice

To determine the causative effect of gut microbiota and PA on T2DM development, we collected feces from the spontaneous T2DM macaques and performed fecal microbiota transplantation (FMT) in antibiotic-pretreated mice. Mice were either administrated by FMT (FT), fed with high concentration PA diet (PA), or were combined FMT and PA diet (FTPA). A control group was used and they were fed with normal commercial food and lacked FMT (**Figure 4A** [↗](#)). FPG monitoring found that FPG levels in the FTPA group and FT group increased continuously from day 60 (**Figure 4B** [↗](#)), while the control group maintained stable FPG levels throughout the 120 days. The FTPA group showed the highest FPG of 6.7 mmol/L at day 120, which was significantly higher than the control group and indicated FTPA-treated mice had developed prediabetes (**Figure 4C** [↗](#)). The oral glucose tolerance test (OGTT) showed obvious glucose intolerance in the FTPA group (**Figures 4D** [↗](#) and **E** [↗](#)), and the FPI values and insulin tolerance test (ITT) were both significantly elevated in the FTPA group, indicating insulin resistance (**Figures 4F** [↗](#) and **G** [↗](#)). Meanwhile, body weight (BW), TC, and TG significantly increased in the FTPA group (**Figures 4H-J** [↗](#)). While mice in the FT group and PA group had significant weight gain and mild insulin resistance, they did not exhibit significant glucose intolerance and no significant elevation in TC and TG levels compared to controls. Histopathological changes in the pancreas and liver of mice were investigated using H-E

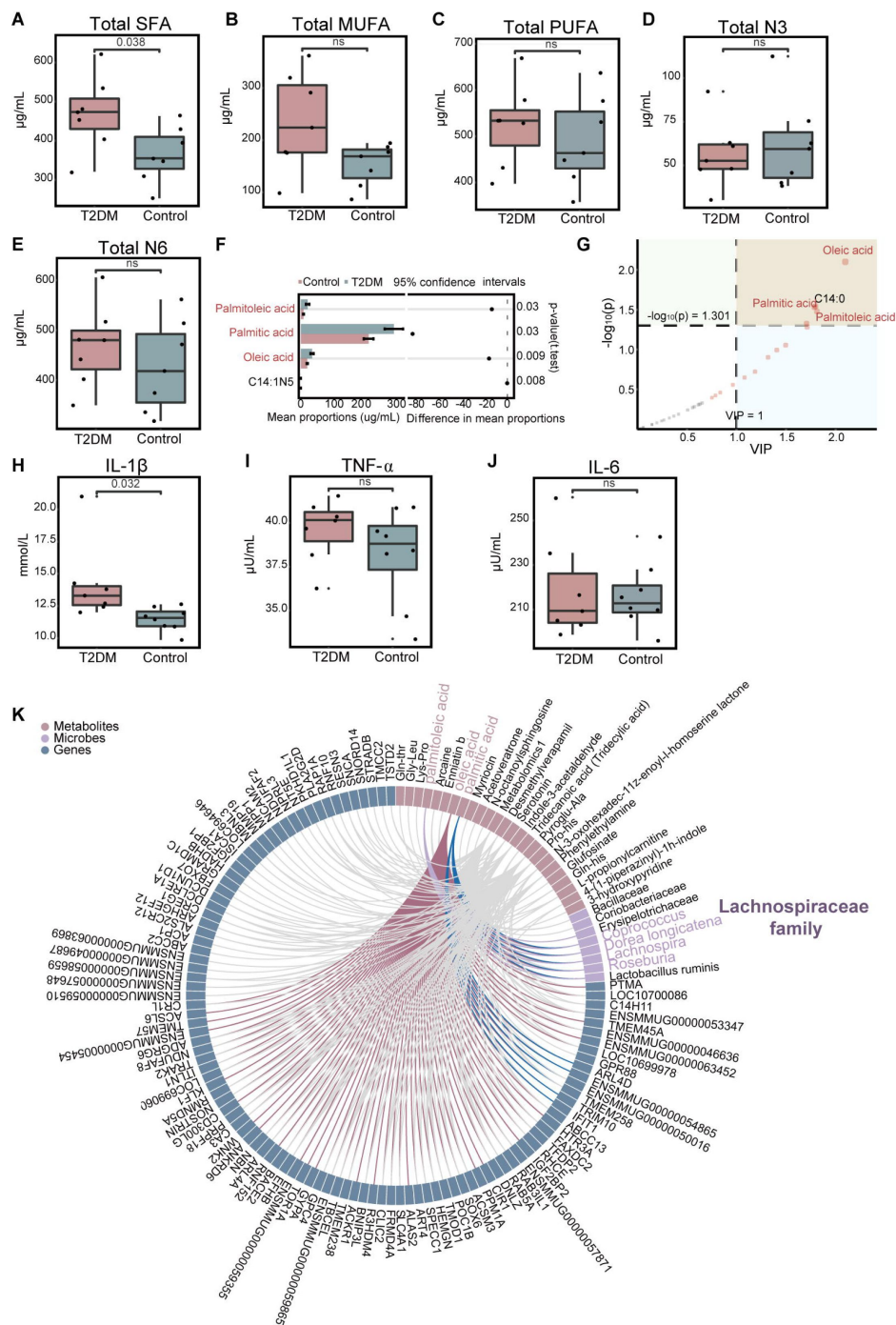


Figure 3.

LCFAs accumulation and inflammation occurred in spontaneous T2DM macaques

(A-E) The contents of SFA (A, $p=0.038$), MUFA (B), PUFA (C), N3 (D), and N6 (E) in plasma (ns, not significant, two-tailed t-test).

(F) The univariate analysis by two-tailed t-test, error bar is mean with $\pm$ s.d.

(G) The multidimensional analysis by VIP value ($VIP>1$).

(H-J) The contents of serum inflammatory cytokines, including IL-1 β (H, $p=0.032$), TNF- α (I) and IL-6 (J) (ns, not significant, two-tailed t-test).

(K) Correlation analysis between DEGs, differential metabolites, and differential microbes using Spearman rank correlation ($|r|>0.5$, $adj\ p<0.05$). For all boxplots: centre lines, upper and lower bounds show median values, 25th and 75th quantiles; upper and lower whiskers show the largest and smallest non-outlier values. Data shown are from 7 individuals per group.

Index	T2DM (n=7)	Control (n=7)
WBC (10e9/L)	15.63±4.66	11.32±2.19*
RBC (10e12/L)	5.47±0.51	5.77±0.43
HGB (g/L)	129.57±15.48	136.71±9.60
HCT (%)	41.31±3.64	44.22±3.12
MCV (fl)	74.13±2.49	76.74±2.00
MCH (pg)	23.17±1.00	23.71±0.60
MCHC (g/L)	312.86±10.16	309±8.14
RDW (%)	15.23±2.12	14.69±1.74
PLT (10e9/L)	402±86.66	371.57±86.42
MPV (fl)	10.54±1.62	10.24±1.12
PCT (%)	0.42±0.08	0.38±0.09
PDW (%)	14.94±0.58	14.83±1.64
LYM% (%)	23.67±10.26	47.71±8.13**
LYM# (10e9/L)	3.46±1.66	5.39±1.52*
MON% (%)	4.18±3.34	6.18±2.29
MON# (10e9/L)	0.65±0.62	0.71±0.38
NEU% (%)	71.13±13.23	44.28±8.96**
NEU# (10e9/L)	11.36±4.91	5.00±1.38**
EOS% (%)	0.91±0.69	1.50±1.94
EOS# (10e9/L)	0.14±0.12	0.17±0.22
BAS% (%)	0.12±0.13	0.33±0.47
BAS# (10e9/L)	0.02±0.02	0.04±0.07
NRBC# (10e9/L)	0	0
NRBC% (%)	0	0

* $p < 0.05$, ** $p < 0.01$

Table 2

blood routine examination of Control and T2DM group

staining. Hepatocytes focal necrosis with inflammatory cell infiltration was commonly observed in the FTPA mice, but not frequent in hepatocytes in FT and PA groups (**Figure 4K**). Furthermore, decreased volume and area in pancreatic islets and inflammatory cell infiltration were detected in the FTPA mice (**Figure 4L**), while such pathological changes were not found in the control group. To eliminate interference from host species divergence in gut microbiota composition, we supplemented mouse experiments using FMT from control macaques (HFT group) (**Figure S4A**). By day 30, the HFT group exhibited significantly lower body weight than the untreated control group ($p < 0.05$) (**Figure S4B**). Throughout the experimental period, FPG levels in both HFT and control groups remained within the normal range (< 6 mmol/L) without significant differences, indicating that transplantation of control macaque microbiota did not induce glycemic alterations (**Figure S4C**).

Specific structure of gut microbiota mediates the absorption of excess PA in the ileum

Given that the content of plasma PA significantly increased in the T2DM macaques (**Figures 3F** and **G**), we examined PA content in the feces, ileum, and serum in mice to compare the prediabetes mice and the spontaneous T2DM macaques. PA content in the serum and ileum of the FTPA group was significantly higher than the control group ($p < 0.05$, **Figures 5A** and **B**), but its content in feces was significantly lower than the control group ($p < 0.05$, **Figure 5C**). This suggested that the absorption of PA was significantly enhanced in the ileum leading to the increase of PA in serum. To verify this conclusion, expression of the *Cd36* gene, a gene involved in the uptake and oxidation of LCFAs (**52**), was examined in the ileum of mice. Ileums showed a significantly upregulated expression of *Cd36* in the FTPA group compared to the control group ($p < 0.05$, **Figure 5D**). In contrast, the level of IL-17A, a protein inhibiting the expression of *Cd36* (**53**), was significantly reduced in the ileum of the FTPA group ($p < 0.05$, **Figure 5E**). Interestingly, the PA group's content of PA, *Cd36* expression and IL-17A was not significantly different from the control group. This indicated that without FMT, the ileum could not absorb PA effectively even when fed in high concentrations. Consequently, gut microbiota mediated the absorption of excess PA in the ileum.

We then compared diversity and composition of microbiota communities between the FTPA, FT, PA and control groups, and a fifth group consisting of the microbiota transplants (TP) from T2DM macaques. Shannon and Simpson indices were lower in the FTPA, FT and PA groups than the control group (**Figures S3A** and **B**). NMDS analysis also indicated a distinct microbiota composition from the control group compared to other groups (NMDS1). In addition, microbiota composition of TP from T2DM macaques was distinct from other groups but a little closer to the FTPA and FT groups (NMDS2, **Figure 5F**). In particular, the Lachnospiraceae family showed the higher abundance in the TP, FTPA and FT groups than in the PA and control groups (**Figure S3C**). The abundances of three members of microbes in Lachnospiraceae (*R. gnavus* (current name: *M. gnavus*), *Coproccoccus* sp. and *Clostridium*) in the FTPA and FT groups gradually increased over time after FMT (**Figure S3D**). The FTPA and FT groups shared many differential microbes compared to the control group, such as the significantly upregulated *R. gnavus* (current name: *M. gnavus*) and *Coproccoccus* sp., and the significantly downregulated Christensenellaceae, F16, *Treponema* sp. and *Fibrobacter succinogenes* (**Figures 5G** and **H**). It is noteworthy that these microbes were also differential microbes between T2DM macaques and controls, and their abundances changed in the same way as in macaques (**Figure 1E**). However, mice in the PA group did not share differential microbes with the spontaneous T2DM macaques. Correspondingly, the change of serum PA content in the PA group was not significantly higher than the control group (**Figure S3E**). Integrating 16S rRNA sequencing data from the HFT, FT, and FTPA groups showed that the antibiotic treatment effectively depleted the gut microbiota, resulting in microbial diversity decreased sharply, with the dominant phyla shifting from Bacteroidota and Bacillota to Pseudomonadota (**Figure S4D-G**). The HFT group restored microbial diversity within 30 days,

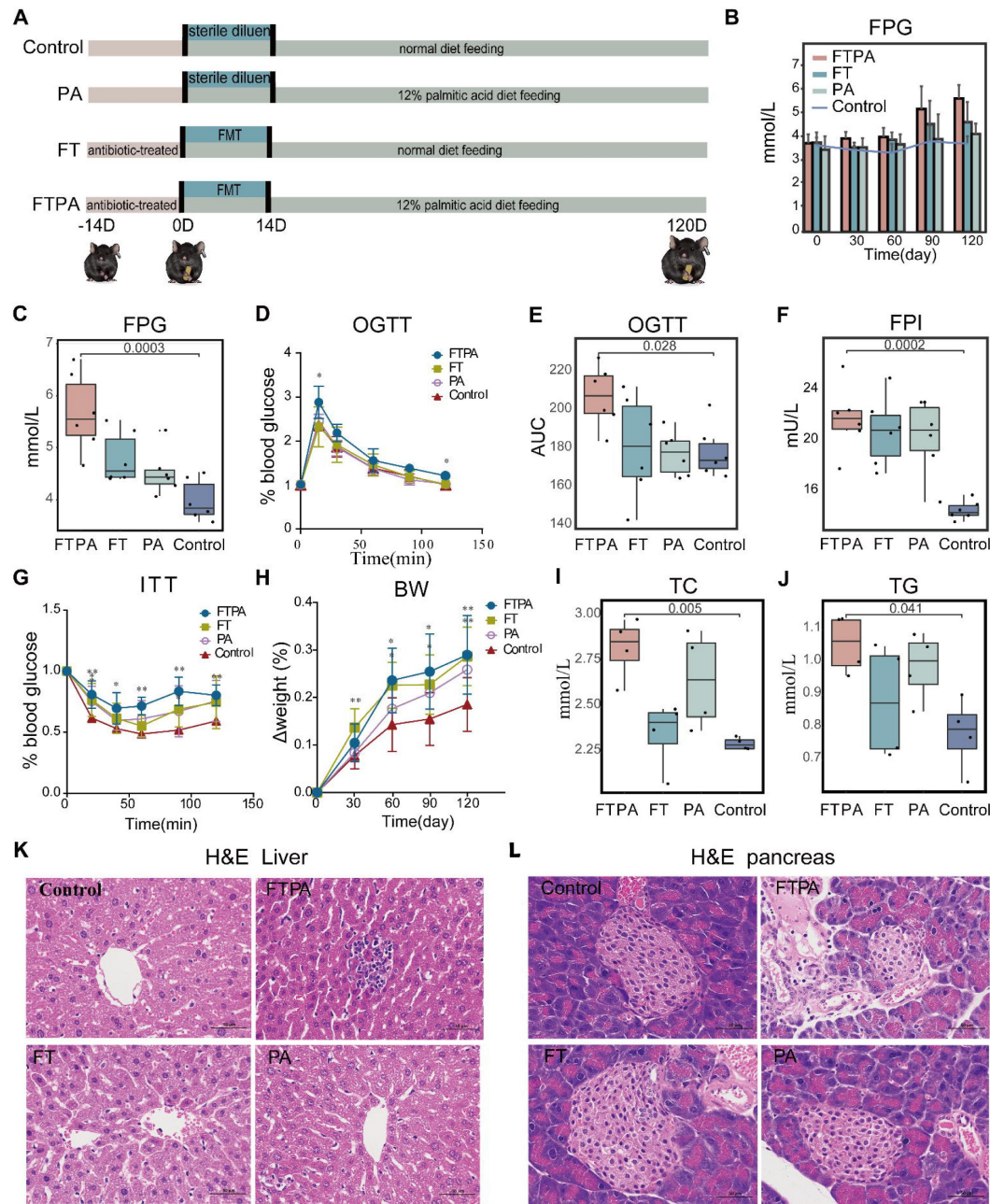


Figure 4.

The FMT and high PA diet mice developed pre-T2DM characteristics

(A) Experimental scheme of FMT and high PA diet treatment.

(B-H) Metabolic analysis, including the trend of FPG within 120 days (B), FPG (C, $p=0.0003$), OGTT (D), AUC of OGTT (E, $p=0.028$), FPI (F, $p=0.007$), ITT (G), and body weight change (H) on day 120.

(I-J) The contents of TC (I, $p=0.005$) and TG (J, $p=0.041$) in serum on day 120.

(K and L) Representative H-E staining images of liver (K) and pancreas (L). For all boxplots: centre lines, upper and lower bounds show median values, 25th and 75th quantiles; upper and lower whiskers show the largest and smallest non-outlier values. Significance was determined using one-way ANOVA. In d, g, and h: $*p<0.05$, $**p<0.01$. Data shown are from 4-6 individuals per group.

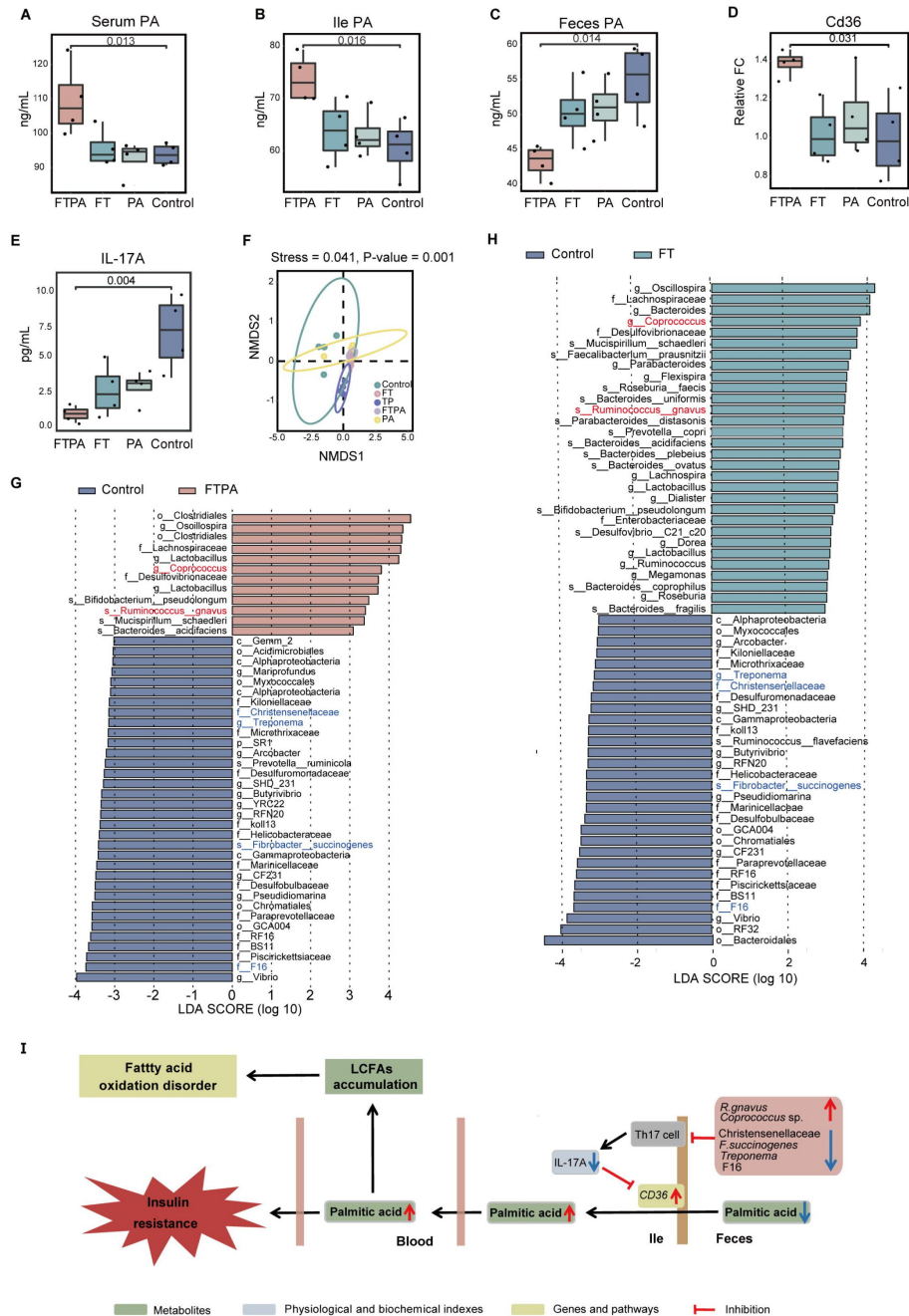


Figure 5.

The PA accumulation required the specific gut microbiota

(A-C) Total PA contents in serum (A, $p=0.013$), ileum (B, $p=0.016$) and feces (C, $p=0.014$) on day 120.

(D) Quantitative RT-PCR for Cd36 transcripts in ileum on day 120 ($p=0.049$).

(E) The content of IL-17A in ileum on day 120 ($p=0.027$). For all boxplots: centre lines, upper and lower bounds show median values, 25th and 75th quantiles; upper and lower whiskers show the largest and smallest non-outlier values. Significance was determined using one-way ANOVA. Data shown are from 3-4 individual macaques per group.

(F) NMDS analysis ($p=0.001$, one-way ANOVA), ellipses represent the 95% confidence intervals.

(G and H) LefSe analysis between FTPA and control groups (G), FT and control groups (H). Data shown are from 4 individuals per group.

(I) Specific gut microbiota structure promoted the absorption of excess PA by regulating the expression of IL-17A and Cd36, leading to the LCFAs accumulation and insulin resistance.

achieving community proportions comparable to untreated controls. Core functional phyla (Bacteroidota and Bacillota) stably colonized in HFT group (Figure S4D-I). Critically, FT and FTPA groups exhibited increased Lachnospiraceae (including genera *Ruminococcus* (current name: *Mediterraneibacter*), *Coproccoccus*, and *Clostridium*) compared with the HFT group on day 30. In addition, LEfSe comparison identified significant *R. gnavus* (current name: *M. gnavus*) enrichment in the FT group (LDA>3, $p<0.01$) (Figure S4J-M). Our results suggested that the transplanted microbiota from spontaneous T2DM macaques, especially the increased abundance of *R. gnavus* (current name: *M. gnavus*) and *Coproccoccus* sp. and decreased abundance of *Treponema*, *F. succinogenes*, Christensenellaceae and F16, promoted the absorption of excess PA by regulating the expression of IL-17A and Cd36, leading to the LCFAs accumulation and insulin resistance (Figures 5I [↗](#)).

Discussion

With a multi-omics technology, this study comprehensively characterizes the gut microbiota, metabolites and gene expression of spontaneous T2DM macaques. The gut microbiota diversity in T2DM macaques decreased. In particular, the abundance of bacteria *R. gnavus* (current name: *M. gnavus*) and Erysipelotrichaceae were upregulated while the abundance of Christensenellaceae was downregulated in the T2DM macaques. Metabolome results demonstrated a decrease of microbiota-derived tryptophan and anti-inflammatory metabolites, indicating that T2DM macaques were prone to inflammation. Notably, the accumulation of acylcarnitine metabolites, the suggested biomarkers for human T2DM ([50 \[↗\]\(#\)](#)), indicated incomplete mitochondrial LCFA β oxidation in T2DM macaques. Transcriptome results identified many DEGs linked to insulin resistance, fatty acid β oxidation and inflammation, including *IGF2BP2*, *LEPR*, *RAP1A*, *SESTRIN 3*, and *ITLN1* that have also been reported in human T2DM ([41 \[↗\]\(#\)](#), [42 \[↗\]\(#\)](#), [43 \[↗\]\(#\)](#), [44 \[↗\]\(#\)](#), [45 \[↗\]\(#\)](#)). Combining the multi-omics results we revealed the complex pathological mechanisms in the spontaneous T2DM macaques (Figure 6 [↗](#)), which is comparable to T2DM humans. Firstly, expression of genes related to lipolysis, fatty acid oxidation, LCFA accumulation, inflammation, and insulin secretion are dysregulated, promoting the development of T2DM (Figure 6A [↗](#)). This is particularly related to lipid metabolism, where the increase of lipolysis and the downregulated expression of fatty acid metabolism-related genes *HADHB* and *ACSM3* cause the accumulation of acylcarnitine and LCFAs, which results in incomplete LCFA oxidation in T2DM macaques (Figure 6B [↗](#)). We also suggest that the decrease of *Lactobacillus* sp. cause the reduction of Aryl hydrocarbon receptor (AhR) ligands (serotonin and indole-3-acetaldehyde) given that *Lactobacillus* is producer of AhR ligands ([54 \[↗\]\(#\)](#)). This then promotes the expansion of the PA producer Erysipelotrichaceae ([55 \[↗\]\(#\)](#), [56 \[↗\]\(#\)](#)) and ultimately leads to the increase of PA levels. The increase in the abundance of Erysipelotrichaceae and *R. gnavus* (current name: *M. gnavus*) also promotes the development of T2DM by activation of inflammation ([57 \[↗\]\(#\)](#), [58 \[↗\]\(#\)](#), [59 \[↗\]\(#\)](#)) (Figure 6C [↗](#)). The accumulation of PA was reported to lead to insulin resistance by affecting genes in the insulin signaling pathway ([16 \[↗\]\(#\)](#)). Our study demonstrates that the significantly changed expression of *RAP1A*, *SESTRIN3*, and *IRS1* in PA-mTORC1-Akt pathway causes insulin resistance in T2DM macaques. Moreover, the increase of PA can promote the development of T2DM by upregulating the NF- κ B signaling pathway (Figure 6D [↗](#)).

We validated the observations from the multi-omics analysis finding the significantly higher inflammatory cytokines IL-1 β and LCFA accumulation, especially significant PA accumulation in the T2DM macaques. And found LCFA metabolites were significantly correlated with bacteria in the Lachnospiraceae family. Numerous studies imply an association of gut microbiota with T2DM development. By transplanting gut microbiota from healthy individuals to T2DM individuals, symptoms such as insulin resistance or inflammation were improved ([6 \[↗\]\(#\)](#), [60 \[↗\]\(#\)](#)). However, there is no evidence to date suggesting gut microbiota have directly causative effects on T2DM development. Our study confirmed the causative effect of gut microbiota and PA on T2DM development by transplanting fecal microbiota from spontaneous T2DM macaques to antibiotic

pretreated mice. We successfully induced prediabetes in mice after combining FMT administration and high PA ingestion. However, when the treatments were administered on their own, the mice did not develop prediabetes. We determined, for the first time, that gut microbiota mediated the absorption of excess PA in the ileum by quantitative examining PA contents in feces, ileum, and serum, and analysis of the expression of *Cd36* and IL-17A level in the ileum. Most notably, this then resulted in accumulation of PA in the serum and finally led to T2DM development. Without the transplanting gut microbiota, the ileum could not absorb the PA effectively even at a high concentration of ingested PA. Our study highlights the essential roles of gut microbiota in T2DM development, which may account for the inability of prior studies to induce T2DM in macaques through high-fat diet intervention alone (28, 29). Furthermore, applying this approach to induce T2DM in macaques will enable deeper investigation into gut-microbiota-driven mechanisms underlying disease pathogenesis.

We then determined the specific gut microbiota structure that related to T2DM development in the prediabetes mice and spontaneous T2DM macaques. We found that the increased abundance of *R. gnavus* (current name: *M. gnavus*) and *Coprococcus* sp., and the decreased abundance of *Treponema*, *F. succinogenes*, Christensenellaceae, and F16, were involved in the T2DM development. *R. gnavus* (current name: *M. gnavus*) can promote insulin resistance by regulating the content of tryptamine/phenethylamine (61). Moreover, *R. gnavus* (current name: *M. gnavus*) is a mucin-degrading microbe that leads to an increase of inflammation (62, 63, 64). The intestinal mucous layer is an important barrier separating intestinal tissue from microbiota, and microbiota composition plays a major role in affecting the integrity of intestinal mucous layer (62). The increase of *R. gnavus* (current name: *M. gnavus*) suggested a higher risk of damage to the integrity of the intestinal mucous layer. This is further supported by our results of the lower abundance of *Lactobacillus* sp., which are AhR ligand producers, and the decreased content of AhR ligands (tryptophan microbial metabolites) in spontaneous T2DM macaques. The deficiency of AhR ligand reduced the production of intestinal mucus and increased the risk of microbial invasion, which in turn affected the immune cell differentiation and cytokine production (65, 66). The cytokine IL-17A is a regulator of the fatty acid transporter *Cd36* and lipid absorption can be promoted by reducing the inhibition of *Cd36* expression by IL-17A (58). The decreased IL-17A level and increased *Cd36* expression in prediabetes mice indicated that the specific gut microbiota promoted the absorption of excess PA by disrupting the integrity of the intestinal mucous layer and regulating the expression of IL-17A and *Cd36*. In addition, the beneficial bacteria decreased in abundance in T2DM macaques and prediabetes mice. These beneficial bacteria, such as *F. succinogenes*, Christensenellaceae, and F16, protect the mucosal barrier and improved insulin resistance (67, 68, 69). We inferred the collectively effects of these gut microbes determined the absorption of excess PA from ileum to serum, which might contribute to the development of T2DM. Previous studies have shown that insulin-resistant patients exhibit increased fecal monosaccharides associated with microbial carbohydrate metabolism (70). Furthermore, commensal species of Lachnospiraceae actively overproduce long-chain fatty acids during metabolic dysfunction through altered bacterial lipid metabolism. The microbe-derived fatty acids impair intestinal epithelial integrity to exacerbate metabolic dysregulation (71). Given that microbial metabolic activity causally modulates host metabolic homeostasis, the content change of PA was potentially associated with a dynamic equilibrium between host absorption and microbial metabolism. Further integrative studies on the fecal fatty acid metabolome, microbial PA metabolism, and functional pathways will be crucial for delineating causal links between dysbiosis and lipid metabolic dysfunction in T2DM.

In conclusion, spontaneous T2DM macaques that have never been treated with diabetes-related drugs provide a valuable model for our understanding of the pathological characteristics and pathogenesis of T2DM. This study characterized changes in gene expression, metabolites, and gut microbiota levels of spontaneous T2DM macaques using multi-omics techniques. We found gut abnormal microbiota, tryptophan metabolism and fatty acid β oxidation disorders, inflammation, and PA accumulation. We also successfully induced prediabetes in mice by transplanting fecal

microbiota from T2DM macaques into antibiotic pretreated mice fed a high PA diet. Our study confirms the causative effect of gut microbiota composition change and PA in T2DM development. The microbiota composition, specifically higher abundance of *R. gnavus* (current name: *M. gnavus*) and *Coprococcus* sp. and lower abundance of *Treponema*, *F. succinogenes*, Christensenellaceae and F16, promoted the absorption of excess PA thus led to the development of T2DM. This study provides new insights into the interaction of gut microbiota and metabolites in the development of T2DM, which expands our understanding of the pathogenesis of this metabolic disease and may provide novel insights for the treatment of T2DM in the future.

Materials and methods

The screening of spontaneous T2DM macaques

The experimental macaques used in this study were all from Greenhouse Biotechnology Co., LTD (Meishan, China). We obtained eight spontaneous T2DM macaques with FPG $\geq$ 7 mmol/L and eight healthy control macaques with FPG $\leq$ 6.1 mmol/L (three consecutive detections, each detection interval of one month) from a population of 1698 captive macaques. None of these 16 screened macaques received any medical treatment for diabetes.

Sample collection

Each experimental macaque was kept in a single cage, and was fasted for 12 hours but had free access to drinking water. We obtained serum, plasma and whole blood for the detection of physiological and biochemical parameters, and metabolome and transcriptome analysis. Fecal samples were collected within ten minutes after deposition. During the sampling process, fecal samples were loaded into a 50 mL sterile centrifuge tube were stored at -80°C . We followed animal welfare guidelines throughout the sample collection process, and all observations and samplings were approved by the Sichuan University's Animal Care Committee.

Physiological and biochemical parameters

In this study, FPG, HbA1c and FPI were detected by hexokinase method, high performance liquid chromatography and electro-chemiluminescence method, respectively. TC, TG, HDL and LDL were detected by automatic biochemical analyzer. HOMA-IR is one of the criteria for T2DM, calculated as: $(\text{FPI} \times \text{FPG}) / 22.5$.

Feces 16S rRNA amplicon sequencing and analysis

Total DNA from fecal samples was extracted using the QIAamp Fast DNA Stool Mini Kit. The V3-V4 region of the 16S rRNA gene was amplified using the 341F/806R primer (341F: 5'-CCTAYGGGRBGCASCAG -3', 806R: 5'-GGACTACNNGGTATCTAAT -3'). The purification and quantification of the amplified products were performed and followed by the sequencing library preparation with TruSeq Nano DNA LT Library Prep Kit (Illumina, USA). The library sequencing was performed on Illumina MiSeq platform and 250 bp paired-end reads were generated. Raw sequencing reads were merged using FLASH (V1.2.7, <http://ccb.jhu.edu/software/FLASH/>) (72) and analyzed by QIIME2 (version 2020.11.1) pipeline with default parameters. Reads after denoising by DADA2 were clustered into OTUs at 99% similarity threshold. Taxonomy of OTUs was assigned based on Greengenes reference database (73). The QIIME2 diversity plugin was used to calculate alpha diversity (74). Principal Coordinate Analysis (PCoA) was determined by using the R package vegan. Differentially microbes were determined using linear discriminant analysis effect size (LEfSe) (75).

Feces shotgun metagenome sequencing and analysis

Total DNA from each sample was extracted using Tiangen DNA Stool Mini Kit (Tiangen Biotech Co., Ltd., China). Metagenome sequencing was performed using the Illumina NovaSeq 6000 platform with a paired-end sequencing length of 150 bp. Trimmomatic was used for removing the adapters and low-quality raw reads based on a four-base wide sliding window, with average quality per base >20 and minimum length 90 bp (76). The rhesus macaque potential sequences were removed using Bowtie2 (77) with the reference genome (assembly Mmul_10). Taxonomy of remaining reads was assigned using Kraken2 (78) with the option “--use-mpa-style”. *De novo* assembly of remaining reads was performed using MEGAHIT (79) with the option “-m 0.90 --min-contig-len 300”. Prodigal (80) was used for gene prediction. The construction of non-redundant gene catalogue was performed with CD-HIT (81) with the option “-c 0.95 -aS 0.90”. Quantification of the non-redundant genes in each sample was performed using Salmon (82). The amino acid sequences translated from non-redundant genes were aligned (--id 80% --query-cover 70% --evalue 1e-5) by DIAMOND (83) in the Carbohydrate-Active enZymes (CAZY) database (84). The annotation metabolic pathway was performed using HUMANN3 (85).

RNA sequencing and DEG analysis

Total RNA was extracted using PAXgene Blood RNA kit. The cDNA Library was constructed following the NEBNext® Ultra™ RNA Library Prep Kit for Illumina® (NEB, USA) manual, and index was added to each sample for sample differentiation. cBot Cluster Generation System was used to cluster the sequences with the same index. Illumina HiSeq 2500 sequencing platform was used to obtain the paired-end sequencing reads (PE150). NGS QC Toolkit v2.3.3 (86) was used to obtain high quality reads (high-quality paired reads with more than 90% of bases with Q-value ≥ 20 were retained). Processed reads of each sample were mapped to the macaque reference genome using HISAT2 v2.1.0 (Kim, et al. 2015). Each alignment output file was assembled into a separate transcriptome using StringTie v1.3.6 (87), resulting in a transcript GTF file. To obtain the expression value of TPM (transcripts per million) and raw read counts of each gene, transcript GTF file was used as the reference annotation file. Differential expression analysis was performed using DESeq2 R package (88). DEGs were screened according to Foldchange value (FC) and p value corrected by FDR (Benjamini-Hochberg method was selected) ($\text{adj } p < 0.05$, $|\log_2 \text{FC}| > 1$).

Weighted Gene Co-Expression Network Analysis (WGCNA) (89) was used to analyse the correlation between genes and phenotypes. We selected an appropriate “soft thresholding power” using the “picksoftthreshold” function in the WGCNA package (v1.61). Next, “blockwiseModules” function was used to construct the co-expression matrix with the option “checkMissingData=TRUE, power=16, TOMType=‘unsigned’, minModuleSize=30, maxBlockSize=6000, mergeCutHeight=0.25”. The modules with high correlation to T2DM phenotype and $p < 0.05$ were selected for downstream analysis, and genes in these modules were used for GO and KEGG functional enrichment analysis. Hub genes were identified based on the module eigengene-based connectivity (kME), $|kME| > 0.8$ as the cut-off criteria.

Instead of identifying the differentially expressed genes within a pathway between the two groups, Aggregate Fold Change (AFC) calculated the average multiple change for each gene and defined the pathway score as the average difference multiple for all genes in that pathway. A null hypothesis test was performed using the pathway scores of the gene expression dataset, and the significance of each pathway was estimated by p-value (90). STRING (v11.0) provides a tool for functional enrichment analysis based on AFC. GO and KEGG enrichment analysis was performed using g: Profiler (91), AFC enrichment analysis was performed using STRING (https://www.string-db.org/cgi/input?sessionId=bfrWW6asD8S4&input_page_show_search=on).

Untargeted metabolomics processing

The 200 μ L homogenized fecal sample was mixed with 800 μ L cold methanol/acetonitrile (1:1, v/v) to remove the protein. The mixture was centrifuged for 15 min (14000g, 4°C) followed by the drying of the supernatant in a vacuum centrifuge. For LC-MS analysis, the samples were re-dissolved in 100 μ L acetonitrile/water (1:1, v/v) solvent. LC-MS/MS analysis was performed using an UHPLC (1290 Infinity LC, Agilent Technologies) coupled to a quadrupole time-of-flight (AB Sciex TripleTOF 6600) in Shanghai Applied Protein Technology Co., Ltd. The samples were separated by Agilent 1290 Infinity LC ultra-high performance liquid chromatography (UHPLC) HILIC column. In both ESI positive and negative modes, the mobile phase contained A=25 mM ammonium acetate and 25 mM ammonium hydroxide in water and B=acetonitrile. The gradient was 85% B for 1 min and was linearly reduced to 65% in 11 min, and then was reduced to 40% in 0.1 min and kept for 4 min, and then increased to 85% in 0.1 min, with a 5 min re-equilibration period employed. AB Triple TOF 6600 mass spectrometer was used to collect the primary and secondary spectra of per sample. After separation by UHPLC, the samples were analyzed by Triple TOF 6600 mass spectrometer (AB SCIEX). Positive and negative modes of electrospray ionization (ESI) were respectively detected.

Targeted medium-and long-chain fatty acid metabolomics processing

A total amount of 100 μ L plasma per sample was taken in 2 mL glass centrifuge tubes and 1mL chloroform methanol solution was added. After 30min of ultrasound, the supernatant was taken and 2mL of 1% sulfuric acid-methanol solution was added. The mixed solution was placed in a water bath at 80°C for 30min, and then 1mL of N-hexane and 5 mL of pure water were added in turn. Next, 500 μ L of supernatant was absorbed, 25 μ L of methyl N-nonaconate was added and mixed. The final sample was detected by GC-MS. All samples were separated by Agilent DB-WAX capillary column (30 m $\times$ 0.25 mm ID $\times$ 0.25 μ m) gas chromatography. Agilent 7890/5975C gas-mass spectrometer was used for mass spectrometry. The chromatographic peak area and retention time were extracted by MSD ChemStation software. The content of medium-and long-chain fatty acids was calculated by drawing a standard curve.

Metabolomics statistical processing

The screening of significant changed metabolites was performed using univariate and multidimensional analysis. Student's t test was applied to determine the significance of differences between two groups. The variable importance in the projection (VIP) value of each variable was obtained from the Orthogonal Partial Least Squares Discriminant Analysis (OPLS-DA) model, which was used to indicate its contribution to the classification. The screening criteria were $p < 0.05$ and $VIP > 1$.

Animal treatments

Specific-pathogen-free C57BL/6J male mice at 6 weeks of age (vendor: Chengdu dossey experimental animals Co., LTD) were randomly assigned to four groups (FTPA, FT, PA and Control). All the mice lived in cages with the same conditions, including 12h light and 12h dark cycles, temperature 22-25°C and humidity 40-60%.

Diets

HPAD was prepared by adding 12% PA to conventional forage. Both the conventional forage and the HPAD were sterile and the fresh forage was renewed three times a week. On day 0, the diets of FTPA-treated mice and PA-treated mice were switched to HPAD, FT-treated mice and control mice were still fed with conventional forage. Drinking water was sterile and renewed twice a day.

Transplant preparation and use

After single cage feeding, FPG detection and fecal collection were performed, and fecal samples of seven T2DM macaques were mixed for the preparation of transplants. The appropriate volume of diluent was added to the fecal sample (i.e. add 4 ml diluent per gram of feces) and the preparation of diluent can be found in (92 [↗](#)). Sodium L-ascorbic acid and L-cysteine hydrochloride monohydrate were added to all suspensions at final concentrations of 5% (w/v) and 0.1% (w/v), respectively (The sterile diluent of control group was added with the same amount of reagent). The mixture was homogenized and filtered with a 200-mesh sterile mesh screen to remove large particles from the feces, and the filtrate was passed through 400 and 800 sterile mesh screens to remove undigested food and smaller particulate matter. The filtrate was divided into 50 ml centrifuge tubes, centrifuged at 600×g for 5 min, and the precipitation was discarded. Finally, the fecal supernatant was divided into new centrifuge tubes in equal parts (400 μL per tube) and frozen at −80°C. For use, the transplant was quickly thawed in a 37 [water bath.

Fecal microbiota transplantation (FMT)

After one week of changed feeding regime, the FTPA and FT mice groups were pre-treated with 1 g/L neomycin sulfate, 1 g/L ampicillin and 1 g/L metronidazole in the drinking water for 14 days, and the control group was not treated. For FMT treatment, the gavage with 400 μL transplant, which were thawed ahead of time, was performed for 14 days in FTPA and FT-treated mice. At the same time, the gavage in control and PA-treated mice were performed with sterile diluent.

Metabolic measurements

Throughout the experiment, body weight and feces were collected every month, FPG was detected every half month under fasting at least 12 h. OGTT was performed on day 110 and ITT was performed on day 115. For OGTT, after a 12h overnight fast, oral glucose gavage (1.2 g/kg of 12% dextrose solution) was performed and followed by blood sample collection from the tail vein at 0, 15, 30, 60, 90 and 120 min. For ITT, after a 6h fast, intraperitoneal insulin injection was performed (0.75 U/kg, human regular insulin), followed by the blood samples collection from the tail vein at 0, 20, 40, 60, 90 and 120 min. On day 120, after FPG detection, blood collection and issues were collected for quantitative RT-PCR, H-E staining and ELISA.

Isolation of tissue for quantitative RT-PCR and ELISA

Tissue was homogenized for RNA extraction, after adjusting the final concentration of RNA, and DNA reverse transcription was performed. Quantitative RT-PCR was performed on CFX Connect Real-Time PCR Detection System (Bio-rad, USA) using SYBR Green (Table 3 [↗](#)-5 [↗](#)). PA, IL-1β, IL-6, TNF-α and IL-17A were detected using Jiangsu Meimian ELISA kit and followed operator instructions.

H-E staining

The isolated livers and pancreas were dehydrated and embedded after fixation with formalin for 24 h. Paraffin-embedded livers and pancreas specimens were cut at a thickness of 3 μm. All sections were stained with hematoxylin then eosin, and finally microscopy and image acquisition were performed.

Statistical analysis

In this study, one-way ANOVA was used to determine statistical significance for comparisons of more than three groups, and for comparisons of two groups, two-tailed t-test was used. p values are represented on figures as follows: ns, not significant, * $p < 0.05$, ** $p < 0.01$.

Table 3**Primers of RT-PCR**

Gene name	Sequences of primers
<i>CD36</i> /F	5'-ATGGGCTGTGATCGGAACTG-3'
<i>CD36</i> /R	5'-GTCTTCCCAATAAGCATGTCTCC-3'
<i>GADPH</i> /F	5'-CCTCGTCCCGTAGACAAAATG-3'
<i>GADPH</i> /R	5'-TCTCCACTTTGCCACTGCAA-3'

Table 4**RT-PCR reaction components**

Reagent	Volume
2×SG Fast qPCR Master Mix	10.0 μL
DNF Buffer	2.0 μL
F primer (10μmol/L)	0.4 μL
R primer (10μmol/L)	0.4 μL
cDNA	1.0 μL
ddH ₂ O	6.2 μL

Table 5**RT-PCR cycle protocol**

Step	Temperature	Time
1	95℃	3 min
2	95℃	1-3 s
3	60℃	30 s
4		
5	72℃	1 min

Data availability

The raw data of transcriptomes, metagenomes, and 16S rRNA have been submitted to the China National Center for Bioinformation/Beijing Institute of Genomics, Chinese Academy of Sciences with BioProject accession no. PRJCA021499. We provide four links that reviewers can use to access the raw data. Transcriptomes data: <https://ngdc.cncb.ac.cn/gsa/s/4xAVJ3Lk> (GSA: CRA013604). 16S rRNA data (macaque): <https://ngdc.cncb.ac.cn/gsa/s/bsDz9qrN> (GSA: CRA013637). 16S rRNA data (mouse): <https://ngdc.cncb.ac.cn/gsa/s/1YcuC464> (GSA: CRA013638). metagenomes data: <https://ngdc.cncb.ac.cn/gsa/s/3k38T849> (GSA: CRA013607). The identified untargeted metabolites were listed in the Table S4. The identified targeted medium-and long-chain fatty acid metabolites were listed in the Table S5.

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Additional information

Author contributions

X.L., G.H., Q.H.L., S.Z.Y. and C.J. collected the samples; X.L. and S.Z.Y. performed the bioinformatics analyses; X.L., Y.C.X., K.S., C.J., J.X.L. and L.Z. performed the experiments; X.L. wrote the manuscript; J.L., Z.L.H., Z.X.F. and B.S.Y. revised the manuscript; J.L. conceived and designed the experiments.

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Additional files

Figure S1. Metagenome analysis of microbiota. (A) Differential microbes screened by metagenome analysis ($p < 0.05$, two-tailed t-test). T2DM: type 2 diabetes mellitus. (B) The proportion of all metabolites. (C) Correlation analysis between differential untargeted metabolites and differential microbes (Spearman's Rho). (D) Correlation analysis between differential targeted metabolites and differential microbes (Spearman's Rho), $*p < 0.05$. Data shown are from 5 individuals per group. [↗](#)

Figure S2. Pathway enrichment analysis by AFC and WGCNA. (A) Total 26 differential pathways were enriched ($FDR < 0.05$). (B) The changes of gene expression in insulin resistance pathway compared to the control group, red color illustrating the up-regulation of genes and blue showed the down-regulation of genes in T2DM group, and the darker the color of the genes, the greater the $|\log_{10}FC|$ value. (C) WGCNA functional enrichment. Data shown are from 8 individuals per group. [↗](#)

Figure S3. The changes in gut microbiota in FTPA, FT, and PA-treated mice. (A) Alpha diversity estimates (Shannon index) ($p > 0.05$, one-way ANOVA, $n = 4$). (B) Alpha diversity estimates (Simpson index) between T2DM and control groups ($p > 0.05$, one-way ANOVA, $n = 4$). (C) Family level taxonomy and relative abundance of five groups. (D) The changes of members of Lachnospiraceae family from day -14 to day 120. (E) LEfSe analysis between PA and control groups. Data shown are from 4 individuals per group. [↗](#)

Figure S4. The changes in gut microbiota in HFT-treated mice. (A) Experimental scheme of fecal FMT from control macaques. (B and C) The physiological characteristic of mice with different treatments, including body weight change (B), and trend of FPG within 30 days (C). * $p < 0.05$. (D-F) Changes in gut microbiota α -diversity and structure in mice, including Shannon index ($p > 0.05$, one-way ANOVA) (D), Simpson index ($p > 0.05$, one-way ANOVA) (E), and NMDS analysis ($p = 0.001$, one-way ANOVA) (F). (G-I) Composition of gut microbiota in mice at phylum level (G), family level (H), and genus level (I). (J and K) The relative abundance of members in Lachnospiraceae between HTP and TP groups (J), control, HFT, FT, and FTPA groups on day 30 (K). (L and M) LEfSe analysis between pre-antibiotic (-14D) and post-antibiotic (0D) groups (L), HFT and FT groups on day 30 (M). [↗](#)

Table S1. Information on the macaque samples. [↗](#)

Table S2. List of 64 differential metabolites. [↗](#)

Table S3. List of genes in four modules significantly correlated with T2DM. [↗](#)

Table S4. List of all identified untargeted metabolites. [↗](#)

Table S5. List of all targeted medium-and long-chain fatty acid metabolites. [↗](#)

References

1. Rayburn W.F (1997) **Diagnosis and classification of diabetes mellitus: highlights from the American Diabetes Association** *J Reprod Med* **42**:585–586 [Google Scholar](#)
2. Zhang L., Chu J., Hao W., Zhang J., Li H., Yang C., Yang J., Chen X., Wang H (2021) **Gut microbiota and type 2 diabetes mellitus: Association, mechanism, and translational applications** *Mediators Inflamm* **17**:e5110276 <https://doi.org/10.1155/2021/5110276> | [Google Scholar](#)
3. Kahn S.E., Cooper M.E., Del Prato S (2014) **Pathophysiology and treatment of type 2 diabetes: perspectives on the past, present, and future** *Lancet* **383**:1068–1083 [https://doi.org/10.1016/s0140-6736\(13\)62154-6](https://doi.org/10.1016/s0140-6736(13)62154-6) | [Google Scholar](#)
4. Kang N., Chen G., Tu R., Liao W., Liu X., Dong X., Li R., Pan M., Yin S., Hu K., et al. (2022) **Adverse associations of different obesity measures and the interactions with long-term exposure to air pollutants with prevalent type 2 diabetes mellitus: The Henan Rural Cohort study** *Environ Res* **207**:112640 <https://doi.org/10.1016/j.envres.2021.112640> | [Google Scholar](#)
5. Qin J., Li Y., Cai Z., Li S., Zhu J., Zhang F., Liang S., Zhang W., Guan Y., Shen D., et al. (2012) **A metagenome-wide association study of gut microbiota in type 2 diabetes** *Nature* **490**:55–60 <https://doi.org/10.1038/nature11450> | [Google Scholar](#)
6. Zhao L., Zhang F., Ding X., Wu G., Lam Y.Y., Wang X., Fu H., Xue X., Lu C., Ma J., et al. (2018) **Gut bacteria selectively promoted by dietary fibers alleviate type 2 diabetes** *Science* **359**:1151–1156 <https://doi.org/10.1126/science.aao5774> | [Google Scholar](#)
7. Karlsson F.H., Tremaroli V., Nookaew I., Bergström G., Behre C.J., Fagerberg B., Nielsen J., Bäckhed F (2013) **Gut metagenome in European women with normal, impaired and diabetic glucose control** *Nature* **498**:99–103 <https://doi.org/10.1038/nature12198> | [Google Scholar](#)
8. Tilg H., Moschen A.R (2014) **Microbiota and diabetes: an evolving relationship** *Gut* **63**:1513–1521 <https://doi.org/10.1136/gutjnl-2014-306928> | [Google Scholar](#)
9. Hendriks T., Schnabl B. (2019) **Indoles: metabolites produced by intestinal bacteria capable of controlling liver disease manifestation** *J. Intern. Med* **286**:32–40 <https://doi.org/10.1111/joim.12892> | [Google Scholar](#)
10. Qi Q., Li J., Yu B., Moon J., Chai J.C., Merino J., Hu J., Ruiz-Canela M., Rebholz C., Wang Z., et al. (2022) **Host and gut microbial tryptophan metabolism and type 2 diabetes: an integrative analysis of host genetics, diet, gut microbiome and circulating metabolites in cohort studies** *Gut* **71**:1098–1105 <https://doi.org/10.1136/gutjnl-2021-324053> | [Google Scholar](#)
11. Wu Z., Zhang B., Chen F., Xia R., Zhu D., Chen B., Lin A., Zheng C., Hou D., Li X., et al. (2023) **Fecal microbiota transplantation reverses insulin resistance in type 2 diabetes: A randomized, controlled, prospective study** *Frontiers in Cellular and Infection Microbiology* **12** <https://doi.org/10.3389/fcimb.2022.1089991> | [Google Scholar](#)

12. Ng S.C., Xu Z., Mak J.W.Y., Yang K., Liu Q., Zuo T., Tang W., Lau L., Lui R.N., Wong S.H., et al. (2021) **Microbiota engraftment after faecal microbiota transplantation in obese subjects with type 2 diabetes: a 24-week, double-blind, randomised controlled trial** *Gut* **71**:716–723 <https://doi.org/10.1136/gutjnl-2020-323617> | Google Scholar
13. Ding D., Yong H., Na Y., Lu W., Xu Y., Ye X., Wang Y., Cai T., Zheng X., Chen H., et al. (2022) **Prospective study reveals host microbial determinants of clinical response to fecal microbiota transplant therapy in type 2 diabetes patients** *Front. Cell. Infect. Microbiol* **12**:820367 <https://doi.org/10.3389/fcimb.2022.820367> | Google Scholar
14. Unger R.H., Zhou Y (2001) **Lipotoxicity of beta-cells in obesity and in other causes of fatty acid spillover** *Diabetes* **50**:S118–S121 <https://doi.org/10.2337/diabetes.50.2007.s118> | Google Scholar
15. Poitout V., Robertson R.P (2008) **Glucolipotoxicity: fuel excess and β -cell dysfunction** *Endocr Rev* **29**:351–366 <https://doi.org/10.1210/er.2007-0023> | Google Scholar
16. Palomer X., Pizarro-Delgado J., Barroso E., Vázquez-Carrera M (2018) **Palmitic and oleic acid: the yin and yang of fatty acids in type 2 diabetes mellitus** *Trends Endocrinol Metab* **29**:178–190 <https://doi.org/10.1016/j.tem.2017.11.009> | Google Scholar
17. Milanski M., Degasperi G., Coope A., Morari J., Denis R., Cintra D.E., Tsukumo D.M.L., Anhe G., Amaral M.E., Takahashi H.K., et al. (2009) **Saturated fatty acids produce an inflammatory response predominantly through the activation of TLR4 signaling in hypothalamus: implications for the pathogenesis of obesity** *J. Neurosci* **29**:359–370 <https://doi.org/10.1523/jneurosci.2760-08.2009> | Google Scholar
18. Fiehn O., Garvey W.T., Newman J.W., Lok K.H., Hoppel C.L., Adams S.H (2010) **Plasma metabolomic profiles reflective of glucose homeostasis in non-diabetic and type 2 diabetic obese African-American women** *PLoS One* **5**:e15234 <https://doi.org/10.1371/journal.pone.0015234> | Google Scholar
19. Li X., Xu Z., Lu X., Yang X., Yin P., Kong H., Yu Y., Xu G (2009) **Comprehensive two-dimensional gas chromatography/time-of-flight mass spectrometry for metabonomics: Biomarker discovery for diabetes mellitus** *Anal Chim Acta* **633**:257–262 <https://doi.org/10.1016/j.aca.2008.11.058> | Google Scholar
20. Chandra K., Jain V., Azhar M., Khan W., Alam O., Ahmad S., Jain S.K (2020) **Effect of augmented glycation in mobilization of plasma free fatty acids in type 2 diabetes mellitus** *Diabetes Metab Syndr* **14**:1385–1389 <https://doi.org/10.1016/j.dsx.2020.07.028> | Google Scholar
21. Ertunc M.E., Hotamisligil G.S (2016) **Lipid signaling and lipotoxicity in metaflammation: indications for metabolic disease pathogenesis and treatment** *J. Lipid Res* **57**:2099–2114 <https://doi.org/10.1194/jlr.R066514> | Google Scholar
22. Salvadó L., Palomer X., Barroso E., Vázquez-Carrera M (2015) **Targeting endoplasmic reticulum stress in insulin resistance** *Trends Endocrinol Metab* **26**:438–448 <https://doi.org/10.1016/j.tem.2015.05.007> | Google Scholar
23. Tumova J., Andel M., Trnka J (2016) **Excess of free fatty acids as a cause of metabolic dysfunction in skeletal muscle** *Physiol Res* **65**:193–207 <https://doi.org/10.33549/physiolres.932993> | Google Scholar

24. Cani P.D., Amar J., Iglesias M.A., Poggi M., Knauf C., Bastelica D., Neyrinck A.M., Fava F., Tuohy K.M., Chabo C., et al. (2007) **Metabolic endotoxemia initiates obesity and insulin resistance** *Diabetes* **56**:1761–1772 <https://doi.org/10.2337/db06-1491> | Google Scholar
25. Birdwell L., Levesque D., Machiah D., Gumber S (2022) **Clinicopathologic characteristics of pancreatic islet amyloidosis in the rhesus macaque (*Macaca mulatta*) and sooty mangabey (*Cercocebus atys*)** *J. Med. Primatol* **51**:155–164 <https://doi.org/10.1111/jmp.12579> | Google Scholar
26. Westermark P., Andersson A., Westermark G.T (2011) **Islet amyloid polypeptide, islet amyloid, and diabetes mellitus** *Physiol. Rev* **91**:795–826 <https://doi.org/10.1152/physrev.00042.2009> | Google Scholar
27. Cefalu W.T (2006) **Animal models of type 2 diabetes: clinical presentation and pathophysiological relevance to the human condition** *Ilar J* **47**:186–198 <https://doi.org/10.1093/ilar.47.3.186> | Google Scholar
28. Ji F., Jin L., Zeng X., Zhang X., Zhang Y., Sun Y., Gao L., He H., Rao J., Liu X., et al. (2012) **Comparison of gene expression between naturally occurring and diet-induced T2DM in cynomolgus monkeys** *Dongwuxue Yanjiu* **33**:79–84 <https://doi.org/10.3724/SP.J.1141.2012.01079> | Google Scholar
29. Tang MT (2020) **Study on the Role of Glucose and Lipid in the Establishment of Type 2 Diabetic Cynomolgus Monkey Model** *M.S. Thesis, Dept. Veterinary Med South China Agricultural Univ* **2020** | Google Scholar
30. Matsuda M., DeFronzo R.A (1999) **Insulin sensitivity indices obtained from oral glucose tolerance testing: comparison with the euglycemic insulin clamp** *Diabetes care* **22**:1462–1470 <https://doi.org/10.2337/diacare.22.9.1462> | Google Scholar
31. Lorenzo C., Hazuda H.P., Haffner S.M (2012) **Insulin resistance and excess risk of diabetes in Mexican-Americans: the San Antonio Heart Study** *J. Clin. Endocr. Metab* **97**:793–799 <https://doi.org/10.1210/jc.2011-2272> | Google Scholar
32. Cowie C.C., Rust K.F., Byrd-Holt D.D., Gregg E.W., Ford E.S., Geiss L.S., Bainbridge K.E., Fradkin J.E (2010) **Prevalence of diabetes and high risk for diabetes using A1C criteria in the US population in 1988–2006** *Diabetes care* **33**:562–568 <https://doi.org/10.2337/dc09-1524> | Google Scholar
33. Yu W., Hao X., Yang F., Ma J., Zhao Y., Li Y., Wang J., Xu H., Chen L., Liu Q., et al. (2019) **Hematological and biochemical parameters for Chinese rhesus macaque** *PLoS One* **14**:e0222338 <https://doi.org/10.1371/journal.pone.0222338> | Google Scholar
34. Du Y., Gu X., Meng H., Aa N., Liu S., Peng C., Ge Y., Yang Z. (2018) **Muscione improves cardiac function in mice after myocardial infarction by alleviating cardiac macrophage-mediated chronic inflammation through inhibition of NF-κB and NLRP3 inflammasome** *Am. J. Transl. Res* **10**:4235–4246 | Google Scholar
35. Alexeev E.E., Lanis J.M., Kao D.J., Campbell E.L., Kelly C.J., Battista K.D., Gerich M.E., Jenkins B.R., Walk S.T., Kominsky D.J., et al. (2018) **Microbiota-derived indole metabolites promote human and murine intestinal homeostasis through regulation of interleukin-10 receptor** *Am. J. Pathol* **188**:1183–1194 <https://doi.org/10.1016/j.ajpath.2018.01.011> | Google Scholar

36. Manzella C., Singhal M., Alrefai W.A., Saksena S., Dudeja P.K., Gill R.K (2018) **Serotonin is an endogenous regulator of intestinal CYP1A1 via AhR** *Sci. Rep* **8**:6103 <https://doi.org/10.1038/s41598-018-24213-5> | Google Scholar
37. Natividad J.M., Agus A., Planchais J., Lamas B., Jarry A.C., Martin R., Michel M., Chong-Nguyen C., Roussel R., Straube M., et al. (2018) **Impaired aryl hydrocarbon receptor ligand production by the gut microbiota is a key factor in metabolic syndrome** *Cell Metab* **28**:737–749 <https://doi.org/10.1016/j.cmet.2018.07.001> | Google Scholar
38. Rutkowski J.M., Knotts T.A., Ono-Moore K.D., McCain C.S., Huang S., Schneider D., Singh S., Adams S.H., Hwang D.H (2014) **Acylcarnitines activate proinflammatory signaling pathways** *Am. J. Physiol. Endocrinol. Metab* **306**:E1378–E1387 <https://doi.org/10.1152/ajpendo.00656.2013> | Google Scholar
39. Tilg H., Moschen A.R (2008) **Inflammatory mechanisms in the regulation of insulin resistance** *Mol. Med* **14**:222–231 <https://doi.org/10.2119/2007-00119.tilg> | Google Scholar
40. Koves T.R., Ussher J.R., Noland R.C., Slentz D., Mosedale M., Ilkayeva O., Bain J., Stevens R., Dyck J.R.B., Newgard C.B., et al. (2008) **Mitochondrial overload and incomplete fatty acid oxidation contribute to skeletal muscle insulin resistance** *Cell Metab* **7**:45–56 <https://doi.org/10.1016/j.cmet.2007.10.013> | Google Scholar
41. Lee Y., Kang E.S., Kim S.H., Han S.J., Kim C.H., Kim H.J., Ahn C.W., Cha B.S., Nam M., Nam C.M., et al. (2008) **Association between polymorphisms in SLC30A8, HHEX, CDKN2A/B, IGF2BP2, FTO, WFS1, CDKAL1, KCNQ1 and type 2 diabetes in the Korean population** *J. Hum. Genet* **53**:991–998 <https://doi.org/10.1007/s10038-008-0341-8> | Google Scholar
42. Su S., Zhang C., Zhang F., Li H., Yang X., Tang X (2016) **The association between leptin receptor gene polymorphisms and type 2 diabetes mellitus: A systematic review and meta-analysis** *Diabetes Res Clin Pract* **121**:49–58 <https://doi.org/10.1016/j.diabres.2016.08.008> | Google Scholar
43. Zhao J (2014) **Molecular mechanisms of AGE/RAGE-mediated fibrosis in the diabetic heart** *World J Diabetes* **5**:860 <https://doi.org/10.4239/wjdv5.i6.860> | Google Scholar
44. Nascimento E.B., Osler M.E., Zierath J.R (2013) **Sestrin 3 regulation in type 2 diabetic patients and its influence on metabolism and differentiation in skeletal muscle** *Am. J. Physiol. Endocrinol. Metab* **305**:E1408–E1414 <https://doi.org/10.1152/ajpendo.00212.2013> | Google Scholar
45. Tabassum R., Mahajan A., Dwivedi O.P., Chauhan G., Spurgeon C.J., Kumar M.V.K., Ghosh S., Madhu S.V., Mathur S.K., Chandak G.R., et al. (2012) **Common variants of SLAMF1 and ITLN1 on 1q21 are associated with type 2 diabetes in Indian population** *Am. J. Hum. Genet* **57**:184–190 <https://doi.org/10.1038/jhg.2011.150> | Google Scholar
46. Junková K., Mirchi L.F., Chylíková B., Janků M., Šilhavý J., Hüttl M., Marková I., Mikláňková D., Včelák J., Malínská H., et al. (2021) **Hepatic transcriptome profiling reveals lack of Acsm3 expression in polydactylous rats with high-fat diet-induced hypertriglyceridemia and visceral fat accumulation** *Nutrients* **13**:1462–1462 <https://doi.org/10.3390/nu13051462> | Google Scholar
47. Sekine Y., Yamamoto K., Kurata M., Honda A., Onishi I., Kinowaki Y., Kawade G., Watabe S., Nomura S., Fukuda S., et al. (2022) **HADHB, a fatty acid beta-oxidation enzyme, is a**

potential prognostic predictor in malignant lymphoma *Pathology* **54**:286–293 <https://doi.org/10.1016/j.pathol.2021.06.119> | [Google Scholar](#)

48. Howson J.M.M., Walker N.M., Smyth D.J., Todd J.A (2009) **Analysis of 19 genes for association with type I diabetes in the Type I Diabetes Genetics Consortium families** *Genes & Immunity* **10**:S74–S84 <https://doi.org/10.1038/gene.2009.96> | [Google Scholar](#)
49. Xu L., Yin L., Jin Y., Qi Y., Han X., Xu Y., Liu K.X., Zhao Y., Peng J. (2020) **Effect and possible mechanisms of dioscin on ameliorating metabolic glycolipid metabolic disorder in type-2-diabetes** *Phytomedicine* **67**:153139 <https://doi.org/10.1016/j.phymed.2019.153139> | [Google Scholar](#)
50. Liu L., Li Y., Guan C., Li K., Wang C., Feng R., Sun C (2010) **Free fatty acid metabolic profile and biomarkers of isolated post-challenge diabetes and type 2 diabetes mellitus based on GC-MS and multivariate statistical analysis** *J. Chromatogr. B* **878**:2817–2825 <https://doi.org/10.1016/j.jchromb.2010.08.035> | [Google Scholar](#)
51. Dinarello C.A., Donath M.Y., Mandrup-Poulsen T (2010) **Role of IL-1 β in type 2 diabetes** *Curr Opin Endocrinol Diabetes Obes* **17**:1 <https://doi.org/10.1097/med.0b013e32833bf6dc> | [Google Scholar](#)
52. Sun J., Su Y., Chen J., Qin D., Xu Y., Chu H., Lu T., Dong J., Qin L., Li W (2023) **Differential roles of CD36 in regulating muscle insulin response depend on palmitic acid load** *Biomedicines* **11**:729 <https://doi.org/10.3390/biomedicines11030729> | [Google Scholar](#)
53. Kawano Y., Edwards M., Huang Y., Bilate A.M., Araujo L.P., Tanoue T., Atarashi K., Ladinsky M.S., Reiner S.L., Wang H.H., et al. (2022) **Microbiota imbalance induced by dietary sugar disrupts immune-mediated protection from metabolic syndrome** *Cell* **185**:3501–3519 <https://doi.org/10.1016/j.cell.2022.08.005> | [Google Scholar](#)
54. Hezaveh K., Shinde R.S., Klötgen A., Halaby M.J., Lamorte S., Ciudad M.T., Quevedo R., Neufeld L., Liu Z.Q., Jin R., et al. (2022) **Tryptophan-derived microbial metabolites activate the aryl hydrocarbon receptor in tumor-associated macrophages to suppress anti-tumor immunity** *Immunity* **55**:324–340 <https://doi.org/10.1016/j.immuni.2022.01.006> | [Google Scholar](#)
55. Brawner K.M., Yeramilli V.A., Duck L.W., Van Der Pol W., Smythies L.E., Morrow C.D., Elson C.O., Martin C.A. (2019) **Depletion of dietary aryl hydrocarbon receptor ligands alters microbiota composition and function** *Sci. Rep* **9**:14724 <https://doi.org/10.1038/s41598-019-51194-w> | [Google Scholar](#)
56. Turpin W., Bedrani L., Espin-Garcia O., Xu W., Silverberg M.S., Smith M.I., Garay J.A.R., Lee S.H., Guttman D.S., Griffiths A., et al. (2020) **Associations of NOD2 polymorphisms with Erysipelotrichaceae in stool of in healthy first degree relatives of Crohn’s disease subjects** *BMC Medical Genet* **21**:204 <https://doi.org/10.1186/s12881-020-01115-w> | [Google Scholar](#)
57. Turnbaugh P.J., Bäckhed F., Fulton L., Gordon J.I (2008) **Diet-induced obesity is linked to marked but reversible alterations in the mouse distal gut microbiome** *Cell Host Microbe* **3**:213–223 <https://doi.org/10.1016/j.chom.2008.02.015> | [Google Scholar](#)
58. Zhang H., DiBaise J.K., Zuccolo A., Kudrna D., Braidotti M., Yu Y., Parameswaran P., Crowell M.D., Wing R., Rittmann B.E., et al. (2009) **Human gut microbiota in obesity and after gastric**

bypass *Proc. Natl. Acad. Sci. U.S.A* **106**:2365–2370 <https://doi.org/10.1073/pnas.0812600106> | [Google Scholar](#)

59. López Almela I., Romaní Pérez M., Bullich Vilarrubias C., Benitez-Páez A., Gómez E.M., Francés R., Liebisch G., Sanz Y. (2021) **Bacteroides uniformis combined with fiber amplifies metabolic and immune benefits in obese mice** *Gut Microbes* **13**:1–20 <https://doi.org/10.1080/19490976.2020.1865706> | [Google Scholar](#)
60. Kootte R.S., Levin E., Salojärvi J., Smits L.P., Hartstra A.V., Udayappan S.D., Hermes G., Bouter K.E., Koopen A.M., Holst J.J., et al. (2017) **Improvement of insulin sensitivity after lean donor feces in metabolic syndrome is driven by baseline intestinal microbiota composition** *Cell Metab* **26**:611–619 <https://doi.org/10.1016/j.cmet.2017.09.008> | [Google Scholar](#)
61. Zhai L., Xiao H., Lin C., Wong H.L.X., Lam Y.Y., Gong M., Wu G., Ning Z., Huang C., Zhang Y., et al. (2023) **Gut microbiota-derived tryptamine and phenethylamine impair insulin sensitivity in metabolic syndrome and irritable bowel syndrome** *Nat. Commun* **14**:4986 <https://doi.org/10.1038/s41467-023-40552-y> | [Google Scholar](#)
62. Paone P., Cani P.D (2020) **Mucus barrier, mucins and gut microbiota: the expected slimy partners?** *Gut* **69**:2232–2243 <https://doi.org/10.1136/gutjnl-2020-322260> | [Google Scholar](#)
63. Png C.W., Lindén S.K., Gilshenan K.S., Zoetendal E.G., McSweeney C.S., Sly L.I., McGuckin M.A., Florin T.H.J (2010) **Mucolytic bacteria with increased prevalence in IBD mucosa augment in vitro utilization of mucin by other bacteria** *Am. J. Gastroenterol* **105**:2420–2428 <https://doi.org/10.1038/ajg.2010.281> | [Google Scholar](#)
64. Henke M.T., Kenny D.J., Cassilly C.D., Vlamakis H., Xavier R.J., Clardy J (2019) **Ruminococcus gnavus, a member of the human gut microbiome associated with Crohn's disease, produces an inflammatory polysaccharide** *Proc. Natl. Acad. Sci. U.S.A* **116**:12672–12677 <https://doi.org/10.1073/pnas.1904099116> | [Google Scholar](#)
65. Yin J., Yang K., Zhou C., Xu P., Xiao W., Yang H (2019) **Aryl hydrocarbon receptor activation alleviates dextran sodium sulfate-induced colitis through enhancing the differentiation of goblet cells** *Biochem Biophys Res Commun* **514**:180–186 <https://doi.org/10.1016/j.bbrc.2019.04.136> | [Google Scholar](#)
66. Ma N., Guo P., Zhang J., He T., Kim S.W., Zhang G., Ma X (2018) **Nutrients mediate intestinal bacteria-mucosal immune crosstalk** *Front. Immunol* **9**:5 <https://doi.org/10.3389/fimmu.2018.00005> | [Google Scholar](#)
67. Oki K., Toyama M., Banno T., Chonan O., Benno Y., Watanabe K (2016) **Comprehensive analysis of the fecal microbiota of healthy Japanese adults reveals a new bacterial lineage associated with a phenotype characterized by a high frequency of bowel movements and a lean body type** *BMC Microbiol* **16**:284 <https://doi.org/10.1186/s12866-016-0898-x> | [Google Scholar](#)
68. Miguel M.A., Lee S.S., Mamuad L.L., Choi Y.J., Jeong C.D., Son A., Cho K.K., Kim E.T., Kim S.B., Lee S.S (2019) **Enhancing butyrate production, ruminal fermentation and microbial population through supplementation with Clostridium saccharobutylicum** *J. Microbiol. Biotechnol* **29**:1083–1095 <https://doi.org/10.4014/jmb.1905.05016> | [Google Scholar](#)
69. Zhong H., Wang J., Abdullah, Hafeez M.A., Guan R., Feng F. (2021) **Lactobacillus plantarum ZJUF2 Prevents High Fat Diet-Induced Insulin Resistance in Association With Modulation**

of the Gut Microbiota *Front. Nutr* **8**:75422 <https://doi.org/10.3389/fnut.2021.754222> | Google Scholar

70. Takeuchi T., Kubota T., Nakanishi Y., Tsugawa H., Suda W., Kwon A.T., Yazaki J., Ikeda K., Nemoto S., Mochizuki T., et al. (2023) **Gut microbial carbohydrate metabolism contributes to insulin resistance** *Nature* **621**:389–395 <https://doi.org/10.1038/s41586-023-06466-x> | Google Scholar
71. Takeuchi T., Kameyama K., Miyauchi E., Nakanishi Y., Kanaya T., Fuji T., Kato T., Sasaki T., Tachibana N., Negishi H., et al. (2023) **Fatty acid overproduction by gut commensal microbiota exacerbates obesity** *Cell metab* **35**:361–375 <https://doi.org/10.1016/j.cmet.2022.12.013> | Google Scholar
72. Magoc T., Salzberg S.L (2011) **FLASH: fast length adjustment of short reads to improve genome assemblies** *Bioinformatics* **27**:2957–2963 <https://doi.org/10.1093/bioinformatics/btr507> | Google Scholar
73. McDonald D., Price M.N., Goodrich J., Nawrocki E.P., DeSantis T.Z., Probst A., Andersen G.L., Knight R., Hugenholtz P (2011) **An improved Greengenes taxonomy with explicit ranks for ecological and evolutionary analyses of bacteria and archaea** *ISME J* **6**:610–618 <https://doi.org/10.1038/ismej.2011.139> | Google Scholar
74. Bolyen E., Rideout J.R., Dillon M.R., Bokulich N.A., Abnet C.C., Al-Ghalith G.A., Alexander H., Alm E.J., Arumugam M., Asnicar F., et al. (2019) **Reproducible, interactive, scalable and extensible microbiome data science using QIIME 2** *Nat. Biotechnol* **37**:852–857 <https://doi.org/10.1038/s41587-019-0209-9> | Google Scholar
75. Segata N., Izard J., Waldron L., Gevers D., Miropolsky L., Garrett W.S., Huttenhower C (2011) **Metagenomic biomarker discovery and explanation** *Genome Biol* **12**:R60 <https://doi.org/10.1186/gb-2011-12-6-r60> | Google Scholar
76. Bolger A.M., Lohse M., Usadel B (2014) **Trimmomatic: a flexible trimmer for Illumina sequence data** *Bioinformatics* **30**:2114–2120 <https://doi.org/10.1093/bioinformatics/btu170> | Google Scholar
77. Langmead B., Salzberg S.L (2012) **Fast gapped-read alignment with Bowtie 2** *Nat. Methods* **9**:357–359 <https://doi.org/10.1038/nmeth.1923> | Google Scholar
78. Wood D.E., Salzberg S.L (2014) **Kraken: ultrafast metagenomic sequence classification using exact alignments** *Genome Biol* **15**:R46 <https://doi.org/10.1186/gb-2014-15-3-r46> | Google Scholar
79. Li D., Liu C.M., Luo R., Sadakane K., Lam T.W (2015) **MEGAHIT: an ultra-fast single-node solution for large and complex metagenomics assembly via succinct de Bruijn graph** *Bioinformatics* **31**:1674–1676 <https://doi.org/10.1093/bioinformatics/btv033> | Google Scholar
80. Hyatt D., Chen G.L., LoCascio P.F., Land M.L., Larimer F.W., Hauser L.J. (2010) **Prodigal: prokaryotic gene recognition and translation initiation site identification** *BMC Bioinform* **11**:119 <https://doi.org/10.1186/1471-2105-11-119> | Google Scholar
81. Fu L., Niu B., Zhu Z., Wu S., Li W. (2012) **CD-HIT: accelerated for clustering the next-generation sequencing data** *Bioinformatics* **28**:3150–3152 <https://doi.org/10.1093/bioinformatics/bts565> | Google Scholar

82. Patro R., Duggal G., Love M.I., Irizarry R.A., Kingsford C (2017) **Salmon provides fast and bias-aware quantification of transcript expression** *Nat. Methods* **14**:417–419 <https://doi.org/10.1038/nmeth.4197> | Google Scholar
83. Buchfink B., Xie C., Huson D.H (2014) **Fast and sensitive protein alignment using DIAMOND** *Nat. Methods* **12**:59–60 <https://doi.org/10.1038/nmeth.3176> | Google Scholar
84. Lombard V., Golaconda Ramulu H., Drula E., Coutinho P.M., Henrissat B (2013) **The carbohydrate-active enzymes database (CAZy) in 2013** *Nucleic Acids Res* **42**:D490–D495 <https://doi.org/10.1093/nar/gkt1178> | Google Scholar
85. Franzosa E.A., McIver L.J., Rahnavard G., Thompson L.R., Schirmer M., Weingart G., Lipson K.S., Knight R., Caporaso J.G., Segata N., et al. (2018) **Species-level functional profiling of metagenomes and metatranscriptomes** *Nat. Methods* **15**:962–968 <https://doi.org/10.1038/s41592-018-0176-y> | Google Scholar
86. Patel R.K., Jain M (2012) **NGS QC Toolkit: a toolkit for quality control of next generation sequencing data** *PLoS One* **7**:e30619 <https://doi.org/10.1371/journal.pone.0030619> | Google Scholar
87. Pertea M., Pertea G.M., Antonescu C.M., Chang T.C., Mendell J.T., Salzberg S.L (2015) **StringTie enables improved reconstruction of a transcriptome from RNA-seq reads** *Nat. Biotechnol* **33**:290–295 <https://doi.org/10.1038/nbt.3122> | Google Scholar
88. Love M.I., Huber W., Anders S. (2014) **Moderated estimation of fold change and dispersion for RNA-seq data with DESeq2** *Genome Biol* **15** <https://doi.org/10.1186/s13059-014-0550-8> | Google Scholar
89. Langfelder P., Horvath S (2008) **WGCNA: an R package for weighted correlation network analysis** *BMC Bioinform* **9**:559 <https://doi.org/10.1186/1471-2105-9-559> | Google Scholar
90. Yu C., Woo H.J., Yu X., Oyama T., Wallqvist A., Reifman J (2017) **A strategy for evaluating pathway analysis methods** *BMC Bioinform* **18** <https://doi.org/10.1186/s12859-017-1866-7> | Google Scholar
91. Raudvere U., Kolberg L., Kuzmin I., Arak T., Adler P., Peterson H., Vilo J (2019) **g:Profiler: a web server for functional enrichment analysis and conversions of gene lists (2019 update)** *Nucleic Acids Res* **47**:W191–W198 <https://doi.org/10.1093/nar/gkz369> | Google Scholar
92. Berland M., Cadiou J., Levenez F., Galleron N., Quinquis B., Thirion F., Gauthier F., Le Chatelier E., Plaza Oñate F., Schwintner C., et al. (2021) **High engraftment capacity of frozen ready-to-use human fecal microbiota transplants assessed in germ-free mice** *Sci. Rep* **11**:4365 <https://doi.org/10.1038/s41598-021-83638-7> | Google Scholar

Author information

Xu Liu

Key Laboratory of Bio-resources and Eco-environment (Ministry of Education), College of Life Sciences, Sichuan University, Chengdu, China

Shengzhi Yang

Key Laboratory of Bio-resources and Eco-environment (Ministry of Education), College of Life Sciences, Sichuan University, Chengdu, China

Yuchen Xie

Key Laboratory of Bio-resources and Eco-environment (Ministry of Education), College of Life Sciences, Sichuan University, Chengdu, China

Cong Jiang

Key Laboratory of Bio-resources and Eco-environment (Ministry of Education), College of Life Sciences, Sichuan University, Chengdu, China

Ke Shang

Key Laboratory of Bio-resources and Eco-environment (Ministry of Education), College of Life Sciences, Sichuan University, Chengdu, China

Jinxia Luo

Key Laboratory of Bio-resources and Eco-environment (Ministry of Education), College of Life Sciences, Sichuan University, Chengdu, China, Sichuan Key Laboratory of Development and Application of Monkey Models for Human Major Disease, Sichuan, China

Lin Zhang

Key Laboratory of Bio-resources and Eco-environment (Ministry of Education), College of Life Sciences, Sichuan University, Chengdu, China

Gang Hu

SCU-SGHB Joint Laboratory on Non-human Primates Research, Meishan, Sichuan, China

Qinghua Liu

SCU-SGHB Joint Laboratory on Non-human Primates Research, Meishan, Sichuan, China

Bisong Yue

Key Laboratory of Bio-resources and Eco-environment (Ministry of Education), College of Life Sciences, Sichuan University, Chengdu, China

Zhenxin Fan

Key Laboratory of Bio-resources and Eco-environment (Ministry of Education), College of Life Sciences, Sichuan University, Chengdu, China

ORCID iD: [0000-0003-0422-9497](https://orcid.org/0000-0003-0422-9497)

Zhanlong He

Institute of Medical Biology, Chinese Academy of Medical Sciences and Peking Union Medical College, Yunnan, China

Jing Li

Key Laboratory of Bio-resources and Eco-environment (Ministry of Education), College of Life Sciences, Sichuan University, Chengdu, China

For correspondence: ljtjf@126.com

Editors

Reviewing Editor

Kiyoshi Takeda

The University of Osaka, Osaka, Japan

Senior Editor

Wendy Garrett

Harvard T.H. Chan School of Public Health, Boston, United States of America

Reviewer #1 (Public review):

Summary:

The authors tried to identify the relationships between gut microbiota, lipid metabolites and the host in type 2 diabetes (T2DM) by using spontaneously developed T2DM in macaques, considered among the best human models.

Strengths:

The authors compared comprehensively the gut microbiota, plasma fatty acids between spontaneous T2DM and the control macaques, and tried verified the results with macaques in high-fat diet-fed mice model.

Weaknesses:

The observed multi-omics on macaques can be done on humans, which weakens the conclusion of the manuscript, unless the observation/data on macaques could cover during the onset of T2DM that would be difficult to obtain from humans.

Regarding the metabolomic analysis on fatty acids, the authors did not include the results obtained from the macaque fecal samples which should be important considering the authors claimed the importance of gut microbiota in the pathogenesis of T2DM. Instead, the authors measured palmitic acid in the mouse model and tried to validate their conclusions with that.

In murine experiments, palmitic acid-containing diet were fed to mice to induce diabetic condition, but this does not mimic spontaneous T2DM in macaques, since the authors did not measure in macaque feces (or at least did not show the data from macaque feces of) palmitic acid or other fatty acids; instead, they assumed from blood metabolome data that palmitic acid would be absorbed from the intestine to affect the host metabolism, and added palmitic acid in the diet in mouse experiments. Here involves the probable leap of logic to support their conclusions and title of the study.

In addition, the authors measured omics data after, but not before, the onset of spontaneous T2DM of macaques. This can reveal microbiota dysbiosis driven purely by disease progression, but does not support the causative effect of gut microbiota on T2DM development that the authors claims.

<https://doi.org/10.7554/eLife.104355.2.sa1>

Author response:

The following is the authors' response to the original reviews

Reviewer #1 (Public review):

Summary:

The authors tried to identify the relationships among the gut microbiota, lipid metabolites, and the host in type 2 diabetes (T2DM) by using macaques that spontaneously develop T2DM, considered one of the best models of the human disease.

Strengths:

The authors comprehensively compared the gut microbiota and plasma fatty acids between macaques with spontaneous T2DM and control macaques and verified the results with macaques on a high-fat diet-fed mice model.

Weaknesses:

Comment 1: The observed multi-omics of the macaques can be done on humans, which weakens the impact of the conclusion of the manuscript.

We fully acknowledge the critical role of human studies in T2DM research. In our study, the spontaneous T2DM macaque model provided a unique window to address inherent challenges in human studies, including medication interference and environmental heterogeneity. Human studies have struggled to standardize confounding factors such as diet, exercise, and antibiotic use. Moreover, most human T2DM patients receive long-term glucose-lowering medications (e.g., metformin), which directly alter gut microbiota composition and function, masking disease-associated microbial signatures (Sun et al., 2018; Petakh et al., 2023). In contrast, the spontaneous T2DM macaques, untreated with glucose-lowering drugs or antibiotics under strictly controlled conditions, revealed microbiota dysbiosis driven purely by disease progression. Our work bridged the gap between rodent studies and human clinical trials, providing an important clinical reference for guiding targeted interventions, particularly microbiota modulation. We sincerely appreciate the valuable comments. We have added background to the part of the introduction, “In fact, T2DM macaques avoid medication interference and environmental heterogeneity under controlled experimental conditions, and share key pathological features with humans, such as amyloidosis of pancreatic islets, which is absent in mouse models (25, 26), suggesting that T2DM macaques are the optimal animal model for simulating human T2DM and its complications (27).” (Lines 98-103).

References:

Sun L., Xie C., Wang G., Wu Y., Wu Q., Wang X., Liu J., Deng Y., Xia J., et al. 2018) Gut microbiota and intestinal FXR mediate the clinical benefits of metformin Nat. Med 24:1919-1929 <https://doi.org/10.1038/s41591-018-0222-4>

Petakh P., Kamyshna I., Kamyshnyi A 2023) Effects of metformin on the gut microbiota: A systematic review Mol. metab 77:101805-101805 <https://doi.org/10.1016/j.molmet.2023.101805>

Comment 2: In addition, the age and sex of the control macaque group did not necessarily match those of the T2DM group, leaving the possibility for compromising the analysis.

Thank you for pointing this out. The availability of spontaneous T2DM macaques is very limited. Wang et al. (2018) identified only nine diabetic macaques among 2,000 screened, and our prior study (Jiang et al., 2022) found merely seven diabetic cases in 1,408 macaques. In this work, we obtained eight spontaneous T2DM macaques with FPG $\geq$ 7 mmol/L and eight healthy control macaques with FPG $\leq$ 6.1 mmol/L (three consecutive detections, each detection

interval of one month) from a population of 1,698 captive macaques. To avoid confound factors affect the investigated macaques, all macaques were individually housed with standardized diets and environmental controls. While age and sex partially matched, controls originated from the same population to minimize confounding. The T2DM and control groups were matched for age period (5 adult and 3 elder) and had comparable mean ages (mean age of T2DM individuals = 12.88, mean age of control individuals = 11.25) (Table S1). In terms of gender matching, we compared blood metabolome data of 12 healthy adult female and 12 healthy adult male macaques from another study (Liu et al., 2023) and obtained only a small number of differential metabolites that were not associated with tryptophan (Table 1). We acknowledge this limitation and will prioritize matched controls in future studies.

Author response table 1.

List of all differential metabolites.

Metabolites	VIP	p.value
L-Proline	2.374869	0.013163
Gamma-Glu-Leu	2.335273	0.016495
2-Aminoethanesulfinic Acid	2.316213	0.019359
N-Methyl-L-Glutamate	2.330681	0.020222
Cis-9,10-epoxystearic acid	2.233362	0.022564
Glycinamide ribonucleotide	2.204234	0.022961
N-(1-Adamantyl)-1-pentyl-1H-indazole-3-carboxamide	2.190427	0.026836
(2S,6R)-6- <u>I</u> [(8S,9S,10R,13R,14S,17R)-10,13-dimethyl-2-oxo-1,2,6,7,8,9,11,12,14,15,16,17-dodecahydrocyclopenta[a]phenanthren-17-yl]-2-methylheptanoate	2.208039	0.028134
Ketamine	2.31113	0.028157
Coniferyl acetate	2.255893	0.028787
2,22-Dideoxy-3-dehydroecdysone	2.191292	0.029986
Myoinositol	2.127819	0.031677
cis-1-Pentadecenoic Acid(C15: 1)	2.136614	0.032026
8-isoprostaglandin F1 α	2.251159	0.033455
9-HpODE	2.164937	0.033538
Carnitine C14:2 Isomer 1	2.11497	0.037547
2-Aminoethanesulfonic Acid	2.094312	0.037956
D-Glucose	2.090356	0.041529
L-Citrulline	2.089761	0.042616
Ethyl dodecanoate	2.142308	0.04289
2-(4-hydroxyphenyl) propionate	2.213272	0.044145
Carnitine C14:3	2.009145	0.045105
3-Sulfocatechol	2.04293	0.046325
Hexadecanedioic acid	2.030341	0.049595

References:

Wang J., Xu S., Gao J., Zhang L., Zhang Z., Yang W., Li Y., Liao S., Zhou H., Liu P., et al. 2018) SILAC-based quantitative proteomic analysis of the livers of spontaneous obese and diabetic rhesus monkeys *Am. J. Physiol-endoc. M* 315:E29-E306 <https://doi.org/10.1152/ajpendo.00016.2018>

Jiang C., Pan X., Luo J., Liu X., Zhang L., Liu Y., Lei G., Hu G., Li J 2022) Alterations in microbiota and metabolites related to spontaneous diabetes and pre-diabetes in rhesus macaques *Genes* 13:1513 <https://doi.org/10.3390/genes13091513>

Liu X., Liu X.Y., Wang X.Q., Shang K., Li J.W., Lan Y., Wang J., Li J., et al. 2023). Multi-Omics Analysis Reveals Changes in Tryptophan and Cholesterol Metabolism before and after Sexual Maturation in Captive Macaques *BMC Genomics* 24:308. <https://doi.org/10.1186/s12864-023-09404-3>

Comment 3: Regarding the metabolomic analysis, the authors did not include fecal samples which are important, considering the authors' claim about the importance of gut microbiota in the pathogenesis of T2DM.

We thank the reviewer for this suggestion. This study employed untargeted metabolomics on macaque fecal samples to identify metabolites associated with spontaneously developing T2DM. To validate the metabolites identified through the untargeted metabolomic analysis, we conducted targeted medium- and long-chain fatty acid (MLCFA) metabolomics on macaque serum, and we further quantitatively examined the content of palmitic acid (PA) in mice feces, ileum, and serum. Although targeted MLCFA metabolomics was not performed on macaque fecal samples, we performed untargeted metabolomics on macaque feces and confirmed the contribution of PA in mice that underwent fecal microbiota transplantation (FMT) from T2DM macaques. We have added future expectations in the part of the discussion, “Previous studies have shown that insulin-resistant patients exhibit increased fecal monosaccharides associated with microbial carbohydrate metabolism (70). Furthermore, commensal species of Lachnospiraceae actively overproduce long-chain fatty acids during metabolic dysfunction through altered bacterial lipid metabolism. The microbe-derived fatty acids impair intestinal epithelial integrity to exacerbate metabolic dysregulation (71). Given that microbial metabolic activity causally modulates host metabolic homeostasis, the content change of PA was potentially associated with a dynamic equilibrium between host absorption and microbial metabolism. Further integrative studies on the fecal fatty acid metabolome, microbial PA metabolism, and functional pathways will be crucial for delineating causal links between dysbiosis and lipid metabolic dysfunction in T2DM.” (Lines 426-437).

Comment 4: In the mouse experiments, the control group should be given a FMT from control macaques rather than just untreated SPF mice since the fecal microbiota composition is likely very different between macaques and mice.

Thanks for your helpful suggestion. We recognized the importance of a FMT control group and supplemented mouse experiments (using the C57BL/6J strain) with FMT from control macaques (HFT group). Another group of mice without FMT was set as control. Due to the lengthy experimental period, observations were concluded at 30 days post-FMT. We compared changes in the gut microbiota before and after antibiotic treatment in mice (-14D and 0D), and tracked body weight and fasting plasma glucose (FPG) levels from day -14 to day 30. At 30 days after FMT, fecal samples from all groups were collected for 16S rRNA sequencing. Additionally, samples of T2DM microbiota transplant (TP), and control transplant (HTP) were sequenced. Finally, we integrated the 16S sequencing data from the FTPA group (palmitic acid (PA) diet and FMT from T2DM macaques) and FT group (normal diet and FMT from T2DM macaques) at day 30 for combined analysis. The results showed that the antibiotic treatment used in this study effectively depleted the gut microbiota. Following FMT, gut

microbial diversity stabilized within 30 days, with similar microbial community proportions between HFT and control groups. Core functional groups of the healthy microbiota (Bacteroidota and Bacillota) stably colonized mice despite host species divergence, confirming that T2DM phenotypes originate specifically from macaque microbiota. Importantly, increased abundance of Lachnospiraceae (including genera *Ruminococcus* (current name: *Mediterraneibacter*), *Coproccoccus*, and *Clostridium*) and the key species *Ruminococcus gnavus* (current name: *Mediterraneibacter gnavus*) were also observed in FT group versus HFT group on day 30, validating our original findings. We have added findings in the results, “To eliminate interference from host species divergence in gut microbiota composition, we supplemented mouse experiments using FMT from control macaques (HFT group) (Figure S4A). By day 30, the HFT group exhibited significantly lower body weight than the untreated control group ($p < 0.05$) (Figure S4B). Throughout the experimental period, FPG levels in both HFT and control groups remained within the normal range (< 6 mmol/L) without significant differences, indicating that transplantation of control macaque microbiota did not induce glycemic alterations (Figure S4C).” (Lines 276-283), and “Integrating 16S rRNA sequencing data from the HFT, FT, and FTPA groups showed that the antibiotic treatment effectively depleted the gut microbiota, resulting in microbial diversity decreased sharply, with the dominant phyla shifting from Bacteroidota and Bacillota to Pseudomonadota (Figure S4D-G). The HFT group restored microbial diversity within 30 days, achieving community proportions comparable to untreated controls. Core functional phyla (Bacteroidota and Bacillota) stably colonized in HFT group (Figure S4D-I). Critically, FT and FTPA groups exhibited increased Lachnospiraceae (including genera *Ruminococcus* (current name: *Mediterraneibacter*), *Coproccoccus*, and *Clostridium*) compared with the HFT group on day 30. In addition, LEfSe comparison identified significant *R. gnavus* (current name: *M. gnavus*) enrichment in the FT group (LDA > 3 , $p < 0.01$) (Figure S4J-M).” (Lines 324-334, 825-837). Specifically:

(1) Experimental design: transplant preparation and FMT from control macaques

After single cage feeding and FPG detection, fecal samples from three control macaques were collected and mixed for transplantation preparation. Then, 4 ml diluent (Berland et al., 2021) was added per gram of feces. Sodium L-ascorbic acid (5% (w/v)) and L-cysteine hydrochloride monohydrate (0.1% (w/v)) were added to all suspensions (The sterile diluent of control group was added with the same amount of reagent). The mixture was homogenized and filtered sequentially through 200, 400, and 800 μ m sterile mesh screens. The filtrate was centrifuged ($600 \times g$, 5 min), and supernatants were aliquoted (400 μ L/tube) for storage at -80°C . For use, the transplant was quickly thawed in a 37°C water bath.

Specific-pathogen-free male C57BL/6J mice aged 6 weeks were randomized into control and HFT (receiving FMT from control macaques) groups. Mice received antibiotic water (ampicillin, neomycin sulfate, and metronidazole, 1 g/L each) from days -14 to 0. All mice were maintained under standard conditions (12h light/dark, $22-25^{\circ}\text{C}$, 40-60% humidity) with sterile diet and twice-daily water changes. Body weight, fasting plasma glucose (FPG) were monitored, and fecal samples were collected throughout the study, with fecal 16S rRNA sequencing performed (Figure S4). The study was approved by the Ethics Committee of College of Life Sciences, Sichuan University, and conducted in accordance with the local legislation and institutional requirements.

(2) Results

Body weight monitoring revealed no significant difference between HFT and control groups before (-14D) and after (0D) antibiotic treatment. By day 30, the HFT group exhibited significantly lower body weight than the untreated control group ($p < 0.05$) (Figure S4B). Throughout the experimental period, FPG levels in both HFT and control groups remained within the normal range (< 6 mmol/L) without significant differences, indicating that

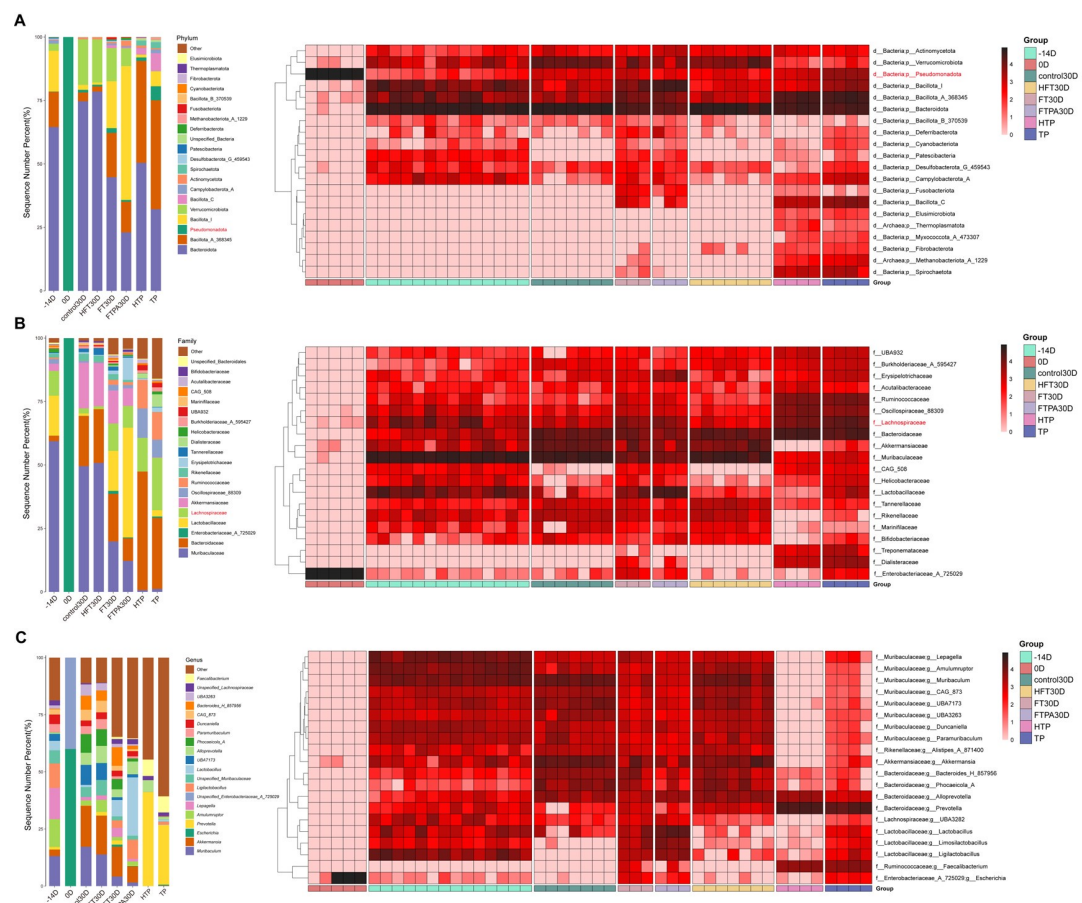
transplantation of control macaque microbiota did not induce glycemic alterations (Figure S4C).

Shannon and Simpson indices showed a significant reduction in gut microbiota diversity after antibiotic treatment (0D) ($p < 0.01$) (Figure S4D,E). The intestinal microbiota of normal mice (-14D) was predominantly composed of Bacteroidota and Bacillota. After two weeks of antibiotic treatment (0D), microbial diversity decreased sharply compared to the -14D group, with the dominant phyla shifting from Bacteroidota and Bacillota to Pseudomonadota (Author response image 1A; Figure S4L). In healthy gut homeostasis, obligate anaerobes such as Bacillota and Bacteroidota maintain intestinal equilibrium. Antibiotic disruption induced dysbiosis in mice, causing substantial restructuring of fecal microbial composition. During dysbiosis, colon epithelial cells shift to anaerobic glycolysis for energy production, increasing epithelial oxygenation and driving expansion of facultative anaerobic Pseudomonadota (de Nies et al., 2023; Szajewska et al., 2024).

NMDS analysis of integrated 16S rRNA sequencing data of FTPA30D (PA diet and FMT from T2DM macaques) and FT30D (normal diet and FMT from T2DM macaques) revealed high intra-group repeatability among pre-antibiotic (-14D), post-antibiotic (0D), HFT30D, T2DM microbiota transplant (TP), and control transplant (HTP) groups. The 0D group showed maximal separation from other clusters, while the -14D, control30D, and HFT30D clustered closely together, with HFT30D nearest to control30D (Figure S4F). On the day 30, all groups showed restoration of microbiota community structure, and the composition of gut microbiota in HFT30D was basically consistent with the control30D group at all taxonomic levels (Author response image 1A-C). At the phylum level, HFT30D group showed significantly reduced relative abundance of Pseudomonadota and increased abundance of Bacteroidota, Bacillota_A, Bacillota_I, and gut barrier-enhancing Verrucomicrobiota (Author response image 1A). These findings demonstrated that FMT from control macaques effectively restored the gut microbiota of antibiotic-treated mice toward a normative state.

Author response image 1.

Composition of gut microbiota in mice. (A) Phylum level; (B) Family level; (C) Genus level.



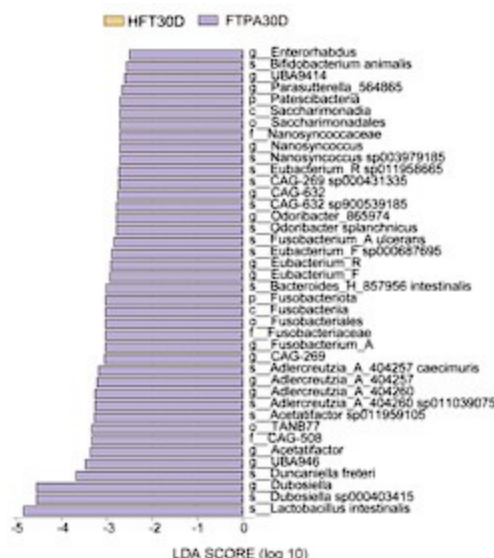
At the phylum level, the FT30D and FTPA30D groups exhibited lower proportions of Bacteroidota/Bacillota compared to the HFT30D (Author response image 1A). Family-level analysis revealed markedly increased abundance of Lactobacillaceae and Lachnospiraceae in FTPA30D and FT30D groups relative to HFT30D, consistent with the changes in the microbiota of spontaneously T2DM macaques (Author response image 1B). Notably, while both HTP and TP groups contained Lachnospiraceae, only FT30D and FTPA30D mice demonstrated significant increase of this family, which was close to that in TP group. Although Muribaculaceae and Bacteroidaceae showed partial recovery in these groups, their relative abundances remained substantially lower than in control30D and HFT30D groups, suggesting that microbiota transplantation from T2DM macaques may reduce specific beneficial taxa while promoting expansion of conditionally pathogenic or metabolically-altered bacteria, such as Lachnospiraceae.

Further analysis of Lachnospiraceae dynamics revealed that at the genus level, most Lachnospiraceae members exhibited higher abundance in the TP group compared to the HTP group. FT30D and FTPA30D groups showed increased abundance of *Ruminococcus* (current name: *Mediterraneibacter*), *Coprococcus*, and *Clostridium* relative to HFT30D group, consistent with prior analyses (Figure S4). LEfSe comparison between FT30D and HFT30D identified significantly enriched *Ruminococcus gnavus* (current name: *Mediterraneibacter gnavus*) in FT30D recipients (LDA > 3, $p < 0.01$), corroborating earlier findings (Figure S4L). As a mucin-degrading microbe, *R. gnavus* (current name: *M. gnavus*) promotes insulin resistance through modulation of tryptamine/phenethylamine levels (Zhai et al., 2023) and exhibits pro-inflammatory properties (Henke et al., 2019; Paone and Cani, 2020). The absence of *R. gnavus* (current name: *M. gnavus*) enrichment in FTPA30D was potentially related to differential

long-term impacts of T2DM microbiota transplantation across the 30- versus 120-day experimental timelines.

Author response image 2.

Identification of differential microbiota in mice. (A) Linear discriminant analysis Effect Size (LEfSe) analysis between pre-antibiotic (-14D) and post-antibiotic (0D) groups; (B) HFT and FTPA groups; (C) HFT and FT groups.



References:

- Berland M., Cadiou J., Levenez F., Galleron N., Quinquis B., Thirion F., Gauthier F., Le Chatelier E., Plaza Oñate F., Schwintner C., et al. 2021) High engraftment capacity of frozen ready-to-use human fecal microbiota transplants assessed in germ-free mice Sci. Rep 11 <https://doi.org/10.1038/s41598-021-83638-7>
- Szajewska H., Scott KP., Meij T de., Forslund-Startceva S.K., Knight R., Koren O., Little P., Johnston B.C., Łukasik J., Suez J., Tancredi D.J., Sanders M.E 2024) Antibiotic-perturbed microbiota and the role of probiotics Nat. Rev. Gastro. Hepat 1-18 <https://doi.org/10.1038/s41575-024-01023-x>
- de Nies L., Kobras C.M., Stracy M 2023) Antibiotic-induced collateral damage to the microbiota and associated infections. Nat. Rev. Microbiol 21:789-804 <https://doi.org/10.1038/s41579-023-00936-9>
- Zhai L., Xiao H., Lin C., Wong H.L.X., Lam Y.Y., Gong M., Wu G., Ning Z., Huang C., Zhang Y., et al. 2023) Gut microbiota-derived tryptamine and phenethylamine impair insulin sensitivity in metabolic syndrome and irritable bowel syndrome Nat. Commun 14 <https://doi.org/10.1038/s41467-023-40552-y>
- Henke M.T., Kenny D.J., Cassilly C.D., Vlamakis H., Xavier R.J., Clardy J 2019) Ruminococcus gnavus, a member of the human gut microbiome associated with Crohn's disease, produces an inflammatory polysaccharide Proc. Nat. Acad. Sci 116:12672-12677 <https://doi.org/10.1073/pnas.1904099116>
- Paone P., Cani P.D 2020) Mucus barrier, mucins and gut microbiota: the expected slimy partners? Gut 69:2232-2243 <https://doi.org/10.1136/gutjnl-2020-322260>

Comment 5: Additionally, the palmitic acid-containing diets fed to mice to induce a diabetes-like condition do not mimic spontaneous T2DM in macaques.

Thanks for your helpful suggestion. We agree that the palmitic acid (PA)-containing diet alone could not fully mimic spontaneous T2DM in macaques. In our study, the PA diet was employed in mouse experiments to investigate whether gut microbiota modulates serum PA levels and mediates T2DM progression. Our critical finding revealed that microbiota was essential for enhanced PA absorption, while simply increasing dietary levels of PA did not effectively enhance intestinal uptake. The fecal microbiota transplantation (FMT) combined with PA-diet approach successfully induced prediabetic states in mice, which can be further applied to the induction of T2DM in macaques. We have added future expectations in the part of the discussion, “Our study highlights the essential roles of gut microbiota in T2DM development, which may account for the inability of prior studies to induce T2DM in macaques through high-fat diet intervention alone (28, 29). Furthermore, applying this approach to induce T2DM in macaques will enable deeper investigation into gut-microbiota-driven mechanisms underlying disease pathogenesis.” (Lines 393-398).

Reviewer #1 (Recommendations for the authors):

General comments

Comment 1: The authors used macaques in this study. The author claims that macaques may be the best animal model to investigate the relationships among gut microbiota, lipid metabolites, and the host in type 2 diabetes (T2DM). However, there have already been some studies investigating these relationships in humans (for example, doi: 10.1016/j.cmet.2022.12.013, and doi: 10.1038/s41586-023-06466-x). The authors should cite and discuss these papers.

We thank the reviewer for this suggestion. We have cited the two papers in the part of discussion, “Previous studies have shown that insulin-resistant patients exhibit increased fecal monosaccharides associated with microbial carbohydrate metabolism (70). Furthermore, commensal species of Lachnospiraceae actively overproduce long-chain fatty acids during metabolic dysfunction through altered bacterial lipid metabolism. The microbe-derived fatty acids impair intestinal epithelial integrity to exacerbate metabolic dysregulation (71).” (Lines 426-432).

Specific comments

Major:

Comment 2: (1) First of all, sex and age of the T2DM and control groups are different (Suppl Table 1). Since the size of the captive population is 1,698, the authors should be able to select the factors including the sex and age of the control group to match those of the T2DM group and they should do so.

In this work, we obtained eight spontaneous T2DM macaques with FPG ≥ 7 mmol/L and eight healthy control macaques with FPG ≤ 6.1 mmol/L (three consecutive detections, each detection interval of one month) from a population of 1,698 captive macaques. To avoid confound factors affect the investigated macaques, all macaques were individually housed with standardized diets and environmental controls. While age and sex partially matched, controls originated from the same population to minimize confounding. The T2DM and control groups were matched for age period (5 adult and 3 elder) and had comparable mean ages (mean age of T2DM individuals = 12.88, mean age of control individuals = 11.25) (Table S1). In terms of gender matching, we compared blood metabolome data of 12 healthy adult female and 12

healthy adult male macaques from another study (Liu et al., 2023) and obtained only a very small number of differential metabolites that were not associated with tryptophan (Author response table 1). We acknowledge this limitation and will prioritize matched controls in future studies.

References:

Liu X., Liu X.Y., Wang X.Q., Shang K., Li J.W., Lan Y., Wang J., Li J., et al. 2023). Multi-Omics Analysis Reveals Changes in Tryptophan and Cholesterol Metabolism before and after Sexual Maturation in Captive Macaques BMC Genomics 24:308. <https://doi.org/10.1186/s12864-023-09404-3>

Comment 3: (2) Are the normal ranges known for the parameters of macaques shown in Table 1? If so, the authors should include those values in Table 1. If not, the authors should show the values of average and SD or SE of all 1,698 individuals as the reference.

We thank the reviewer for this suggestion. In this study, the normal ranges of fasting plasma glucose (FPG), fasting plasma insulin (FPI), homeostasis model assessment- insulin resistance (HOMA-IR), and glycosylated hemoglobin A1c (HbA1c) were referenced against human standards. According to the American Diabetes Association (ADA) for glucose metabolism status and the diagnostic criteria for diabetes, individuals with FPG ≥ 7 mmol/L were diagnosed as T2DM subjects, and individuals with FPG ≤ 6.1 mmol/L were controls. More sensitive assays show a normal fasting plasma insulin level to be under 12 μ U/mL (Matsuda and DeFronzo, 1999). HOMA-IR ≥ 2.67 indicated the possibility of insulin resistance, which is used in clinical diagnosis (Lorenzo et al., 2012). HbA1c percentages higher than 6.5% were used as an auxiliary diagnostic index for diabetic macaques (Cowie et al., 2010). The normal ranges of triglycerides (TG), total cholesterol (TC), high-density lipoprotein cholesterol (HDL), and low-density lipoprotein cholesterol (LDL) were referenced against the blood lipid index of rhesus macaques (Yu et al., 2019). We have added the normal ranges of parameters to Table 1, “FPG: fasting plasma glucose (normal range: ≤ 6.1 mmol/L); FPI: fasting plasma insulin (normal range: ≤ 12 μ U/mL); HOMA-IR: homeostasis model assessment- insulin resistance (normal range: ≤ 2.67); BMI: body mass index; HbA1c: glycosylated hemoglobin A1c (normal range: $< 6.5\%$); TG: triglycerides (normal range: 0.95 ± 0.47 mmol/L); TC: total cholesterol (normal range: 3.06 ± 0.98 mmol/L); HDL: high-density lipoprotein cholesterol (normal range: 1.62 ± 0.46 mmol/L); LDL: low-density lipoprotein cholesterol (normal range: 2.47 ± 0.98 mmol/L). (30, 31, 32, 33).”.

References:

Matsuda M., DeFronzo R.A 1999) Insulin sensitivity indices obtained from oral glucose tolerance testing: comparison with the euglycemic insulin clamp Diabetes care 22:1462-1470 <https://doi.org/10.2337/diacare.22.9.1462>

Lorenzo C., Hazuda H.P., Haffner S.M 2012) Insulin resistance and excess risk of diabetes in Mexican-Americans: the San Antonio Heart Study J. Clin. Endocr. Metab 97:793-799 <https://doi.org/10.1210/jc.2011-2272>

Cowie C.C., Rust K.F., Byrd-Holt D.D., Gregg E.W., Ford E.S., Geiss L.S., Bainbridge K.E., Fradkin J.E 2010) Prevalence of diabetes and high risk for diabetes using A1C criteria in the US population in 1988–2006 Diabetes care 33:562-568 <https://doi.org/10.2337/dc09-1524>

Yu W., Hao X., Yang F., Ma J., Zhao Y., Li Y., Wang J., Xu H., Chen L., Liu Q., et al. 2019) Hematological and biochemical parameters for Chinese rhesus macaque PLoS One 14:e0222338 <https://doi.org/10.1371/journal.pone.0222338>

Comment 4: (3) The authors measured the fasting plasma glucose (FPG) levels, but it is common to measure whole blood glucose since glucose is consumed during the processing of obtaining plasma which could compromise the results. Please explain why plasma glucose levels were measured.

The criteria for screening spontaneous T2DM macaques were guided by the American Diabetes Association (ADA) for glucose metabolism status and the diagnostic criteria for diabetes. Individuals with FPG ≥ 7 mmol/L were diagnosed as T2DM subjects, and individuals with FPG ≤ 6.1 mmol/L were controls. For the identified subjects, a total of three times of FPG tests were employed, with an interval of one month to reduce the possible error. These individuals were raised in a single cage, and blood samples were collected after an overnight fast at least 12 h. After the three test results meet the standards, venous blood was collected for FPG testing to ensure the reliability of the data to the greatest extent. We have added FPG values of three time to the Table S1.

Comment 5: (4) Since the BMI of the T2DM and control groups did not significantly differ ($p > 0.05$, Table 1), the food intake of the two groups may not significantly differ as well. The authors should examine the food intake data. The food intake is also important in considering the relevance of feeding the PA diet in mice experiments. Were the intake of T2DM macaques including PA more than the control group?

All macaques in this study were individually housed under standardized environments with timed and measured feeding to minimize confounders. Given the non-significant BMI difference between T2DM and control groups, food intake was probably not significantly different. In this study, our findings highlight the essential roles of gut microbiota in T2DM development, and this is probable also the reason that previous studies have failed to induce T2DM in macaques because they have only used a high-fat diet (Ji et al., 2012; Tang, 2020). We agree that PA intake in T2DM macaques warrants focused investigation. Future investigations will incorporate detailed dietary monitoring including palmitic acid (PA) intake and nutrient composition to examine potential relationships between specific dietary components, metabolic parameters, and diabetes progression.

References

Ji F., Jin L., Zeng X., Zhang X., Zhang Y., Sun Y., Gao L., He H., Rao J., Liu X., et al. 2012) Comparison of gene expression between naturally occurring and diet-induced T2DM in cynomolgus monkeys *Dongwuxue Yanjiu* 33:79–84 <https://doi.org/10.3724/SP.J.1141.2012.01079>

Tang MT. 2020) Study on the Role of Glucose and Lipid in the Establishment of Type 2 Diabetic Cynomolgus Monkey Model M.S. Thesis, Dept. Veterinary Med., South China Agricultural Univ. 2020

Comment 6: (5) It may be that the fecal microbiome of the T2DM macaques is involved in the pathogenesis of T2DM; however, it is more important how the gut microbiota compositions were obtained/established by those T2DM macaques. There was no description of when the fecal samples were collected during the course of T2DM. If it was after T2DM symptoms appeared, the authors should perform gut metagenome and also gut metabolome analyses to see the change in those parameters to try to understand how gut microbiome changes are induced leading to T2DM pathogenesis.

The spontaneous T2DM macaques untreated with glucose-lowering drugs or antibiotics, revealed microbiota dysbiosis driven purely by disease progression. After macaques met diagnostic thresholds across three FPG assessments (each detection interval of one month),

we collected fresh fecal samples and stored them aseptically at -80 °C until analysis. The scarcity of spontaneous T2DM macaques precludes invasive sampling, restricting tissue collection to naturally deceased diabetic individuals, which prevented us to explicitly define the disease stage of the T2DM individuals. We recognize the scientific value of gut metagenomic and metabolomic analyses to track microbiome evolution during diabetes progression. This study explored the interaction of gut microbiota and metabolites in T2DM macaques, and future studies can continue to investigate its dynamic changes in the disease process of T2DM.

Comment 7: (6) Regarding the fatty acids, the authors only measured them in the plasma, but they also should measure in feces, since the authors focus on gut microbiota; in addition, a recent report showed fecal fatty acids, especially elaidic acid, contributed the pathogenesis of obesity and T2DM by acting on the gut epithelial cells (doi: 10.1016/j.cmet.2022.12.013). Besides, this study showed the link between a Lachnospiraceae species and fecal palmitic and elaidic acids, which the authors also focused on in this manuscript.

We thank the reviewer for this suggestion. This study employed untargeted metabolomics on macaque fecal samples to identify metabolites associated with spontaneously developing T2DM. To validate the metabolites identified through the untargeted metabolomic analysis, we conducted targeted medium- and long-chain fatty acid (MLCFA) metabolomics on macaque serum, and we further quantitatively examined the content of palmitic acid (PA) in mice feces, ileum, and serum. Although targeted MLCFA metabolomics was not performed on macaque fecal samples, we did perform untargeted metabolomics on macaque feces and confirmed the contribution of PA in mice that underwent fecal microbiota transplantation (FMT) from T2DM macaques. We have added future expectations in the part of the discussion, “Previous studies have shown that insulin-resistant individuals exhibit increased fecal monosaccharides associated with microbial carbohydrate metabolism (70). Furthermore, commensal species of Lachnospiraceae actively overproduce long-chain fatty acids during metabolic dysfunction through altered bacterial lipid metabolism. The microbe-derived fatty acids impair intestinal epithelial integrity to exacerbate metabolic dysregulation (71). Given that microbial metabolic activity causally modulates host metabolic homeostasis, the content change of PA was potentially associated with a dynamic equilibrium between host absorption and microbial metabolism. Further integrative studies on the fecal fatty acid metabolome, microbial PA metabolism, and functional pathways will be crucial for delineating causal links between dysbiosis and lipid metabolic dysfunction in T2DM.” (Lines 426-437).

Comment 8: (7) In FMT and PA diet experiments, SPF mice were used as the control group. However, the gut microbiota composition of the SPF mice is markedly different from that of macaques; the difference must be much bigger than the difference between T2DM and healthy control macaques; therefore, mice with FMT from healthy control macaques have to be used as the control group. As mentioned above (in point #4), is the feeding of mice with PA diet a relevant model reflecting the condition observed in macaques in this study?

Thanks for your helpful suggestion. We recognized the importance of a FMT control group and supplemented mouse experiments (using the C57BL/6J strain) with FMT from control macaques (HFT group). Another group of mice without FMT was set as control. Due to the lengthy experimental period, observations were concluded at 30 days post-FMT. We compared changes in the gut microbiota before and after antibiotic treatment in mice (-14D and 0D), and tracked body weight and fasting plasma glucose (FPG) levels from day -14 to day 30. At 30 days after FMT, fecal samples from all groups were collected for 16S rRNA sequencing. Additionally, samples of T2DM microbiota transplant (TP), and control transplant (HTP) were sequenced. Finally, we integrated the 16S sequencing data from the FTPA group

(palmitic acid (PA) diet and FMT from T2DM macaques) and FT group (normal diet and FMT from T2DM macaques) at day 30 for combined analysis. The results showed that the antibiotic treatment used in this study effectively depleted the gut microbiota. Following FMT, gut microbial diversity stabilized within 30 days, with similar microbial community proportions between HFT and control groups. Core functional groups of the healthy microbiota (Bacteroidota and Bacillota) stably colonized mice despite host species divergence, confirming that T2DM phenotypes originate specifically from macaque microbiota. Importantly, increased abundance of Lachnospiraceae (including genera *Ruminococcus* (current name: *Mediterraneibacter*), *Coproccoccus*, and *Clostridium*) and the key species *Ruminococcus gnavus* (current name: *Mediterraneibacter gnavus*) were also observed in FT group versus HFT group on day 30, validating our original findings. We have added findings in the results, “To eliminate interference from host species divergence in gut microbiota composition, we supplemented mouse experiments using FMT from control macaques (HFT group) (Figure S4A). By day 30, the HFT group exhibited significantly lower body weight than the untreated control group ($p < 0.05$) (Figure S4B). Throughout the experimental period, FPG levels in both HFT and control groups remained within the normal range (< 6 mmol/L) without significant differences, indicating that transplantation of control macaque microbiota did not induce glycemic alterations (Figure S4C).” (Lines 276-283), and “Integrating 16S rRNA sequencing data from the HFT, FT, and FTPA groups showed that the antibiotic treatment effectively depleted the gut microbiota, resulting in microbial diversity decreased sharply, with the dominant phyla shifting from Bacteroidota and Bacillota to Pseudomonadota (Figure S4D-G). The HFT group restored microbial diversity within 30 days, achieving community proportions comparable to untreated controls. Core functional phyla (Bacteroidota and Bacillota) stably colonized in HFT group (Figure S4D-I). Critically, FT and FTPA groups exhibited increased Lachnospiraceae (including genera *Ruminococcus* (current name: *Mediterraneibacter*), *Coproccoccus*, and *Clostridium*) compared with the HFT group on day 30. In addition, LEfSe comparison identified significant *R. gnavus* (current name: *M. gnavus*) enrichment in the FT group (LDA > 3 , $p < 0.01$) (Figure S4J-M).” (Lines 324-334, 825-837).

We agree that the PA-containing diet alone could not fully mimic spontaneous T2DM in macaques. In our study, the PA diet was employed in mouse experiments to investigate whether gut microbiota modulates serum PA levels and mediates T2DM progression. Our critical finding revealed that microbiota was essential for enhanced PA absorption, while simply increasing dietary levels of PA did not effectively enhance intestinal uptake. The FMT combined with PA-diet approach successfully induced prediabetic states in mice, which can be further applied to the induction of T2DM in macaques. We have added future expectations in the part of the discussion, “Our study highlights the essential roles of gut microbiota in T2DM development, which may account for the inability of prior studies to induce T2DM in macaques through high-fat diet intervention alone (28, 29). Furthermore, applying this approach to induce T2DM in macaques will enable deeper investigation into gut-microbiota-driven mechanisms underlying disease pathogenesis.” (Lines 393-398).

Comment 9: FPG was measured here in the mouse experiments, but there was no description of whether mice were under fasting conditions, and this should be clarified. If there are no fasting durations, this should be described in the Materials and Methods section.

As suggested, we have added description to the Materials and Methods section, “Throughout the experiment, body weight and feces were collected every month, FPG was detected every half month under fasting at least 12 h.” (Lines 619-620).

Comment 10: From the PA contents in feces, ileum, and serum in mice (Figures 5A-D), the authors concluded that the absorption of PA was significantly enhanced in the ileum

leading to the increase of PA in serum. However, it could also be possible that consumption of PA by gut microbiota occurs at the same time and the authors should discuss the possibility.

We thank the reviewer for spotting this. We have added a discussion to the manuscript, “Previous studies have shown that insulin-resistant individuals exhibit increased fecal monosaccharides associated with microbial carbohydrate metabolism (70). Furthermore, commensal species of Lachnospiraceae actively overproduce long-chain fatty acids during metabolic dysfunction through altered bacterial lipid metabolism. The microbe-derived fatty acids impair intestinal epithelial integrity to exacerbate metabolic dysregulation (71). Given that microbial metabolic activity causally modulates host metabolic homeostasis, the content change of PA was potentially associated with a dynamic equilibrium between host absorption and microbial metabolism. Further integrative studies on the fecal fatty acid metabolome, microbial PA metabolism, and functional pathways will be crucial for delineating causal links between dysbiosis and lipid metabolic dysfunction in T2DM.” (Lines 426-437).

Comment 11: (8) Nomenclature and classification of bacteria has been revised by the List of Prokaryotic names with Standing in Nomenclature (LPSN) (<https://psn.dsmz.de/>) and recognized as Global Core Biodata Resource in 2023. For example, Ruminococcus gnavus is now Mediterraneibacter gnavus. Therefore, the name of microbes should be corrected accordingly; one proposal is to show the revised correct name with the previous name in parenthesis, such as "Mediterraneibacter gnavus (previously Ruminococcus gnavus)".

Thank you for pointing this out. We have corrected the name of microbe, “*Ruminococcus* (current name: *Mediterraneibacter*)”, “*Ruminococcus gnavus* (current name: *Mediterraneibacter gnavus*)”, and “*R. gnavus* (current name: *M. gnavus*)” (Lines 146, 313, 316-317, 336, 345, 367-368, 401, 404-405, 409, 448, 764-765)

Minor:

Comment 12:

(1) The sentence starting "A total of..." (lines 143-144) seems grammatically wrong; a word such as "represented" should be inserted after "differentially", or alternatively "differentially" should be "differential"?

(2) "medium-and" (line 220) needs a space between "medium-" and "and" to make it "medium- and".

(3) Abbreviations should be spelled out when they appear for the first time in the main text; for example, WBC, NEU, and LYM in line 237.

(4) Should FGP (line 437) be FPG?

(5) What is the definition of "prediabetes" in mice? Is this clearly defined elsewhere?

We sincerely thank the reviewer for careful reading. As suggested, we have improved the statements and revised it according to the requirements:

(1) Line 143: “A total of 21 microbes were identified as differential microbes”.

(2) Line 221: “targeted medium- and long-chain fatty acid”.

(3) Lines 238-239: “white blood cell (WBC)”, “neutrophil (NEU)”, and “lymphocyte (LYM)”.

(4) Line 472: “FPG, HbA1c and FPI were detected”.

(5) Prediabetes or impaired glucose regulation (IGR) is diagnosed when one exhibits blood glucose level higher than normal yet below the diabetic threshold, which is even more prevalent than T2DM in the population (American Diabetes, 2021). Given the higher glycemic diagnostic criteria in mice, we assessed diabetic manifestations integrating physiological and pathological evidence. Compared to control mice, those receiving FMT from T2DM macaques combined with a high-palmitic-acid diet (FTPA group) developed prediabetic characteristics by day 120. Physiological alterations included elevated fasting plasma glucose (FPG), increased fasting plasma insulin (FPI), impaired glucose tolerance, heightened insulin resistance, weight gain, and elevated serum total cholesterol (TC) and triglyceride (TG) levels. Particularly in pathological changes, hepatocytes focal necrosis with inflammatory cell infiltration was commonly observed in FTPA group, alongside decreased volume in pancreatic islets and inflammatory cell infiltration (lines 258-276).

References:

American Diabetes Association 2021) 2. Classification and diagnosis of diabetes: standards of medical care in diabetes—2021 Diabetes care 44:S15-S33 <https://doi.org/10.2337/dc21-S002>

Reviewer #2 (Public review):

This study analyzes the interaction among the gut microbiota, lipid metabolism, and the host in type 2 diabetes (T2DM) using rhesus macaques. The authors first identified 8 macaques with T2DM from 1698 individuals. Then, they observed in T2DM macaques: dysbiosis by 16S rRNA gene amplicon analysis and shotgun sequencing, imbalanced tryptophan metabolism and fatty acid beta oxidation in the feces by metabolome analysis, increased plasma concentration of palmitic acid by MS analysis, and an inflammatory gene signature of blood cells by transcriptomic analysis. Finally, they transplanted feces of T2DM macaques into mice and fed them with palmitic acid and showed that those mice became diabetic through increased absorption of palmitic acid in the ileum.

Comment 1: This study clearly shows the interaction among gut microbiota, lipid metabolism, and the host in T2DM. The experiments were well designed and performed, and the data are convincing. One point I would suggest is that in the experiments of mice with FMT, control mice should be those colonized with feces of healthy macaques, but not with no FMT.

See response to Reviewer 1, Public review comment 4.

<https://doi.org/10.7554/eLife.104355.2.sa0>